

Sustainable Personalised Driving: Applying Human-in-the-Loop Multi-Objective Bayesian Optimisation to Eco-Driving Interface Design

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Abstract

Eco-driving displays are usually evaluated with a few fixed designs identical for all drivers. This study treated display design as a per-participant multi-objective search problem. Sixteen participants each drove 21 short urban rounds in a virtual-reality simulator: 15 Sobol sampling rounds, five proposed by a human-in-the-loop multi-objective Bayesian optimiser over seven continuous display parameters, and one repeat of a recommended configuration. The objectives were a simulated energy-intensity proxy, task load, and three acceptance ratings. In this fixed-order, single-session design, cumulative hypervolume increased for 15 of 16 participants, and the proportion of rounds on a participant's own Pareto front was 0.44 during optimisation versus 0.24 during sampling. The recommended configurations improved all four subjective objectives relative to a same-route sampling baseline, left the energy proxy unchanged, and differed markedly between participants. These fixed-order associations do not establish that personalisation outperforms a strong fixed default.

CCS Concepts

• **Human-centered computing** → **Human computer interaction (HCI)**; • **Computing methodologies** → *Machine learning*.

Keywords

eco-driving, head-up display, human-in-the-loop optimisation, multi-objective Bayesian optimisation, personalisation, virtual reality

1 Introduction

Road transport is one of the largest and most difficult-to-reduce sources of carbon emissions within the transport sector, and the energy used by the same vehicle on the same road depends substantially on how it is driven [2, 19]. Eco-driving - a set of behaviours including gentle acceleration, anticipatory braking, and maintaining a steady speed - is therefore a relatively low-cost solution: it requires neither a new powertrain nor new infrastructure [2]. In practice, in-vehicle feedback is one strategy for promoting it.

The evidence that feedback is effective is substantial. A meta-analysis of in-vehicle eco-driving feedback combined 23 effect sizes from 17 controlled studies and reported a pooled improvement in fuel economy of 6.6%, with a 95% confidence interval of 4.9% to 8.2% [18].

How feedback is designed matters as much as whether it exists. A study of 23 lorry drivers in a moving-base simulator compared continuous and intermittent presentation of visual eco-driving advice and recommended intermittent presentation because it produced shorter glance durations and allowed designers to control where increases in glance frequency occurred [15]. Whether a display expresses information numerically or symbolically changes fuel consumption measured in the field [7]. Neither the amount of information nor the modality through which it is provided has an intuitively predictable effect. In a paired comparison of 12 candidate interfaces, a visual display that provided more information - the required rate of change in pedal angle in addition to current fuel consumption - reduced driver control error. An accompanying auditory cue improved performance and further reduced visual distraction [12]. A design framework for eco-driving feedback categorises information, timing, and presentation as three design dimensions and relates them respectively to the behavioural mechanisms of attention, motivation, and learning [17]. Motivational content based on comparisons with other drivers is intuitively appealing, but did not produce more economical driving when tested [6].

Previous research has generally presented the same display to all participants and compared a small number of predefined conditions. Such designs can estimate average condition effects, but are less able to determine which parameter combination should be used for a particular driver. Earlier work shows that drivers respond differently to goal setting, message framing, and anchoring [20], and personalised eco-driving feedback has also been explored directly [3]. Sumner et al. [21], meanwhile, inferred latent cognitive constructs from recent driving behaviour and used them to trigger a safety interface. Together, these precedents suggest that individual differences should be incorporated into the design process.

Human-in-the-loop Bayesian optimisation offers one candidate method. It treats user ratings and system measurements as expensive, noisy observations of a design space and uses them to select the next design for evaluation [4, 10]. The method has recently been applied to multi-objective automotive-interface optimisation, including visualisations of automated-driving functions, external vehicle communication, and in-vehicle conversational assistants [5, 14, 22]. Eco-driving displays are distinctive because the system must handle both a simulated energy proxy calculated from the driving trajectory and participants' subjective acceptance; the two types of objective differ in their measurement chains, noise, and perceptibility.

Motivated by this background, this study explores a specific approach: whether an eco-driving display can be treated as a human-in-the-loop multi-objective search problem that proposes candidate display configurations online for each driver within a single session. I therefore built an immersive virtual-reality driving simulator reproducing neighbourhoods around Knightsbridge, Belgravia, and Chelsea in central London, together with a parameterised eco-driving display. Each participant completed 21 rounds on short routes and rated the display after every round; the system used their previous ratings and simulated energy-intensity proxy values to propose subsequent designs.

This study addresses three research questions.

RQ1: In a single session with sampling fixed before optimisation, how do multi-objective quality indicators change between the two phases, and what internal comparison is obtained between the five optimisation rounds and the preceding five sampling rounds?

RQ2: How much do the resulting personalised designs differ between drivers, and which parts of the display show the greatest differentiation?

RQ3: Is a 21-round, single-session optimisation protocol feasible and acceptable to participants in an immersive simulator?

RQ1 is a fixed-order within-session comparison. The study did not include an independent non-adaptive condition, nor were sampling and optimisation rounds randomised or interleaved. Later changes therefore cannot be attributed solely to the optimiser, and the study cannot establish that personalisation is superior to a generic default display. Comparing the five optimisation rounds with the preceding five sampling rounds provides only a budget-matched internal reference and remains confounded by time, practice, rating drift, and different search paths.

Under the contribution types proposed by Wobbrock and Kientz [25], this study primarily makes two contributions. The first is empirical material: a study of per-participant, multi-objective search for eco-driving feedback that reports optimisation indicators across fixed-order phases, participant-associated design differentiation, and experiences of the protocol. The second is an artefact contribution: a parameterised display, self-authored components of a virtual-reality driving simulator, and an implementation connecting them to a multi-objective optimiser. The optimisation algorithm is adopted from previous work; this study does not claim a new optimisation method.

2 Related Work

This section reviews four related strands. It first summarises the effect and magnitude of in-vehicle eco-driving feedback, then examines how information, timing, and presentation constitute a display-design space. It next reviews differences in driver needs and existing personalised approaches, and finally introduces human-in-the-loop multi-objective optimisation as the basis for the research motivation.

2.1 Eco-Driving and the Effect of In-Vehicle Feedback

Eco-driving refers to a set of driving behaviours that reduce the energy required for a vehicle to complete a given journey, principally gentle acceleration, anticipatory deceleration, maintaining a steady

speed, and reducing idling. Barkenbus [2] argues that eco-driving is undervalued as a climate measure precisely because it does not require fleet replacement or new infrastructure, only a change in how existing vehicles are driven. Sivak and Schoettle [19] decompose drivers' influence on fuel economy into strategic decisions (vehicle choice and maintenance), tactical decisions (route choice and load), and operational decisions (driving behaviour). They conclude that vehicle choice has a far larger effect than any other factor, with the least and most fuel-efficient models on sale differing ninefold. The remaining factors under the driver's control can nevertheless together produce approximately 45% variation in real-world fuel consumption. In-vehicle feedback acts on this final level.

The evidence that feedback is effective is substantial. Sanguinetti et al. [18] synthesised 23 effect sizes from 17 controlled studies and reported a pooled improvement in fuel economy of 6.6%, 95% CI [4.9%, 8.2%], $p < .0001$. Two of their moderator findings are directly relevant to design: feedback reporting both instantaneous and cumulative performance had a larger effect, and intervention duration was negatively associated with effect size, meaning that benefits diminished as drivers lived with the display for longer. A review of eco-driving guidance for human-driven vehicles reached a similar conclusion while emphasising that whether such systems are used depends not only on effectiveness but also on acceptability [23]. Reported effect sizes vary substantially between studies, and part of that variance should be attributed to the interventions themselves.

This final point motivates the remainder of this section: the effect of eco-driving feedback depends on how the display is designed.

2.2 The Design Space of Eco-Driving Displays

Sanguinetti et al. [17] proposed a framework that organises eco-driving feedback design along three axes: what information is displayed, when it is displayed, and how it is displayed. The value of the framework here is that it explicitly identifies these as independent design decisions, each with its own evidence base.

With respect to timing, Kircher et al. [15] compared four conditions with 23 lorry drivers in a moving-base lorry simulator: a baseline with no eco-driving information, intermittent feedback, continuous feedback, and a personalised condition configured by the driver. Drivers looked at the system when it was active and looked more often when traffic conditions were easier; in more complex situations, they delayed their first glance for longer. The authors recommended intermittent presentation for two reasons: it produced shorter glance durations, and it allowed designers to control where increases in glance frequency occurred by scheduling information where it would not burden the driver. The study already included a condition in which drivers chose their own settings, an early form of the problem addressed here.

With respect to presentation, Dahlinger et al. [7] conducted a randomised controlled field trial comparing numerical and symbolic eco-driving feedback and measured differences in actual fuel consumption. The comparison matters to this study because it isolated representational form: both displays carried the same underlying information and differed only in how that information was drawn.

The trial involved 62 roadside-assistance drivers in Switzerland, lasted ten weeks, covered more than 245,000 km, and analysed 265,939 dynamic road segments with constant road characteristics.

Participants were randomly assigned to three groups. The symbolic group ($N = 23$) saw an 'eco-tree' that grew or withered according to a three-minute moving average of fuel consumption. The numerical group ($N = 24$) saw a fuel-consumption gauge presenting real-time consumption with a needle and number, together with brake- and accelerator-pedal positions. The control group ($N = 25$) retained only a basic screen without colour coding or acceleration information. Fixed-effects regressions found that only symbolic feedback significantly reduced fuel consumption, by approximately 0.2–0.3 L/100 km, or 2%–3%, and that this result was robust across models controlling for road type, speed limit, gradient, time of day, and other covariates. Numerical feedback was consistently non-significant. One detail is worth noting: in the simplest model containing only the treatment variable, the coefficient for numerical feedback was significantly positive, appearing to increase fuel consumption; it became non-significant only after date was included as a covariate.

Regarding layout and integration with the driving task, Jamson et al. [12] used paired comparisons in a high-fidelity driving simulator to evaluate 12 candidate eco-driving interfaces. The system gave real-time advice on the most economical accelerator-pedal angle through an instrument-panel visual display, a multimodal combination of visual and auditory cues, or a haptic accelerator pedal, while varying presentation style within each modality. Effects were measured using both subjective ratings and the difference between the pedal angle requested by the system and that actually applied by the driver. Contrary to the intuition that more information is necessarily more distracting, a visual display that provided the required rate of change in pedal angle in addition to current fuel consumption reduced drivers' pedal error. Adding an auditory cue both improved eco-driving performance and reduced visual distraction from the road ahead. Among the six haptic designs, force feedback - a step-like resistance from the pedal during inefficient acceleration - outperformed stiffness feedback.

Regarding motivational content, Colley et al. [6] added peer comparison to a windscreen eco-driving feedback display and evaluated it in a within-participant virtual-reality study with 19 participants across manual and automated driving modes. Peer comparison is appealing because social comparison is an established motivational mechanism, but the result was null: adding the comparison element did not produce significantly more economical driving in either mode. What mattered was the driver's own score, which produced the clearest improvement in eco-driving; displaying the scores of other vehicles neither improved driving style nor significantly worsened performance or subjective ratings. Their symbolic windscreen layout and eco-score construction were the direct starting points for the display used here, while their null finding is why the final design does not include aggregate peer comparison.

Presentation modality is another axis, and one result on this axis should temper the assumption that a visual eco-driving display is helpful by default. Pietra et al. [16] built a virtual-reality driving simulator with a steering wheel, pedals, and head-mounted display and compared haptic feedback through the accelerator pedal with visual feedback on a virtual head-up display (HUD) in ten participants, both separately and in combination. Relative to a no-feedback condition, haptic feedback reduced their mean power index by 14%, the combined visual-haptic condition reduced it by 11%, and visual

feedback alone increased it by 11%. This was a small study, and I do not treat it as definitive, but it is the closest published setting to the present one and suggests that visual eco-driving feedback in virtual reality can worsen energy use.

Taken together, these studies establish three points: eco-driving displays occupy a large design space; different points within that space produce measurably different outcomes; and trade-offs exist between those outcomes. They do not establish which point in the space should be presented to a particular driver.

2.3 Individual Differences and Personalised Feedback

The studies above compare a small number of predefined designs across samples of drivers and report average differences between them. Such studies answer which display is best on average. They cannot answer whether the display that is best on average is also the best display for a particular driver, and there are good reasons to expect that it often is not.

Stillwater and Kurani [20] interviewed 46 plug-in hybrid drivers in a multi-week field trial with feedback-off (baseline) and feedback-on (treatment) phases. Three quarters of the drivers changed how they drove after feedback was introduced, at least in the short term. Two findings are directly relevant here. First, 'personalization of information allows different drivers to access information meaningful to them, thus increasing its motivational value'. Second, instantaneous performance feedback, such as real-time efficiency or power, was used primarily to experiment with and learn new eco-driving behaviours, whereas average-performance feedback was used primarily to set and achieve goals. The study relied on self-report, and quantitative fuel-consumption effects were left to a subsequent paper. Brouwer et al. [3] directly explored personalised eco-driving feedback and provide the closest precedent for the present question.

That study used a within-participant design with 26 professional lorry drivers in the TNO truck-driving simulator. Each experienced four conditions: a baseline without the system, a basic display, and customised displays for drivers categorised as either learning-oriented or performance-oriented on the basis of value-orientation and goal-orientation theories. The results were reported as group means and were directionally consistent with the personalisation hypothesis: learning-oriented drivers rated both customised displays as more useful than the basic display, whereas performance-oriented drivers considered the learning-oriented display the least acceptable and least useful. The authors were cautious, explicitly describing the increase in acceptance among learning-oriented drivers as an 'increase that could not be validated', and acknowledged that the sample was underpowered. Each session lasted almost four hours, and they estimated that an adequate between-group test would require at least 80 participants. A further point is directly relevant to the energy result in this study: they deliberately excluded simulated fuel consumption from the analysis because 'it is difficult to obtain realistic fuel consumption data in a driving simulator'. The study therefore supports the premise that personalisation may be needed, but does not establish whether personalisation is effective for energy use.

More recent work has personalised driver-facing interfaces by first inferring characteristics of the individual. Sumner et al. [21]

trained a recurrent neural network to infer two latent cognitive constructs - impulsivity and inhibitory control - from recent driving behaviour and used them to trigger a driver-facing safety interface. This approach is powerful when a reliable inference model exists, but introduces an additional dependency: the system can personalise only along dimensions represented by its driver model, and building that model requires independent data collection and validation.

An alternative is to omit the driver model entirely and optimise the interface directly from the driver's own responses. This is what human-in-the-loop optimisation does.

2.4 Human-in-the-Loop Optimisation of User Interfaces

Human-in-the-loop optimisation treats the interface design space as a search space and users' responses as observations of an unknown objective function over that space. Bayesian optimisation is well suited to this setting because it was designed for objectives that are expensive and noisy to evaluate, which precisely describes asking a person to use an interface and rate it [10]. A surrogate model, usually a Gaussian process, is first fitted to the existing observations; an acquisition function then selects the next design for evaluation by balancing exploration of uncertain regions against exploitation of promising regions.

Chan et al. [4] investigated the strengths and weaknesses of human-in-the-loop optimisation in a controlled experiment in which 40 novice designers optimised a three-dimensional touch-interaction technique. The optimiser helped designers cover a larger proportion of the design space, obtain better solutions, and reduce their mental effort. It also had a cost: optimiser-guided designers reported lower agency and expressiveness, felt less creative, and felt less able to control the process. The authors therefore concluded that the method was suitable where agency was not critical. This trade-off is directly relevant here because my participants rated designs selected for them and had no control over the sequence.

When several outcomes matter simultaneously and cannot be combined into a single score without arbitrary weights, the problem becomes multi-objective. Multi-objective Bayesian optimisation returns a Pareto set in which no design can improve one objective without worsening another. Daulton et al. [8] proposed a differentiable expected hypervolume improvement acquisition function, making this process tractable for parallel and batch evaluations, and implemented it in the BoTorch framework [1]. Hypervolume - the volume of objective space dominated by the current Pareto set relative to a reference point - provides a single scalar summary of the quality of the overall trade-off surface, which is why progress can be tracked across iterations.

This mechanism has recently been applied to automotive interfaces. Jansen et al. [14] optimised visualisations of automated-driving functions against six user-experience objectives with user participation in a between-participant online study of 117 participants. They compared three human-in-the-loop conditions - one cold start and two warm starts - against a design produced by eight automotive-interface experts, a design manually constructed by

each end user, and a no-visualisation control. The optimisation conditions significantly outperformed the expert and user-generated designs in perceived safety, trust, and predictability, and produced lower cognitive load; no significant effects were found for acceptance or aesthetics. The null result for acceptance is worth noting because three of the five objectives in the present study measure acceptance. Colley et al. [5] applied the same method to external communication by automated vehicles, and Susak et al. [22] extended it to an in-vehicle conversational assistant with multiple competing objectives.

OptiCarVis also reported an observation directly relevant to the present design space [14]. Under optimisation, many parameters were pushed towards being switched off, in marked contrast to expert designs, whereas this rarely occurred when users configured the display manually. An optimiser given the option to remove elements will use it, and the present parameterisation - in which an element disappears when its size is reduced to zero - provides exactly that option.

Two points in this strand are notable for eco-driving. First, all objectives optimised in these three studies were subjective, or measured through participants' judgements. Eco-driving feedback differs because the environmentally consequential objective is the energy consumed by the driver, which is measured by the vehicle and can be affected by the driver only indirectly through behaviour. Second, these studies did not address eco-driving and did not need to reconcile an objective measure of behaviour change with subjective measures of acceptance within the same optimisation loop.

2.5 Summary and Research Motivation

The literature supports four relatively robust observations. First, in-vehicle eco-driving feedback can improve fuel economy on average [18, 23]. Second, its effect depends on design decisions concerning information, timing, and presentation [7, 12, 15, 17]. Third, drivers differ in how they interpret and respond to the same feedback, and personalisation has been explored directly [3, 20, 21]. Fourth, human-in-the-loop multi-objective Bayesian optimisation has been used to search automotive-interface design spaces under limited user-evaluation budgets [4, 5, 14, 22].

Together, these studies raise a question that still requires methodological exploration: how to adapt an eco-driving display online with the driver in the loop while considering both an energy objective calculated from the driving trajectory and subjective acceptance. Prior research has separately addressed personalised eco-driving feedback, human-in-the-loop automotive-interface optimisation, and objective driving outcomes; this study explores their combination as an implementation to be tested.

Accordingly, this study examines changes in multi-objective indicators within a fixed evaluation budget (RQ1), differentiation between the resulting participant configurations and its possible sources (RQ2), and the feasibility and experience of a 21-round virtual-reality protocol (RQ3). The research questions and conclusions are limited by the fixed phase order, absence of a strong fixed baseline, and use of a simulated energy proxy.

3 Method

3.1 Study Design

The study used a within-participant design. Each participant experienced a sequence of display designs from the same design space, but the sequence was determined by a per-participant optimisation loop rather than by a fixed condition order. Each participant completed 21 driving rounds. The first 15 rounds formed a quasi-random space-filling phase, the next five were proposed by a multi-objective Bayesian optimiser fitted to that participant’s own observations, and the final round evaluated the design recommended by the optimiser at the end of the session.

The independent variable controlled by the optimiser was the display configuration, a point in a seven-dimensional continuous design space. Five dependent variables were returned to the optimiser as objectives: energy use for that round, one task-load score, and three independent acceptance ratings. Energy use and task load were minimised, while the three acceptance ratings were maximised.

The study did not include an independent non-adaptive baseline, and the sampling phase was fixed before the optimisation phase. The results can therefore describe only associations across fixed-order phases and budget-matched internal comparisons within the same design space and participant; they cannot identify an independent causal effect of the optimiser relative to time, practice, drift in questionnaire responses, or differences between search paths.

3.2 Participants

Sixteen participants took part (P1–P16). They were 22–25 years old ($M = 23.1$, $SD = 0.9$) and had held a driving licence for 2–7 years ($M = 3.6$, $SD = 1.5$); ten were men, five were women, and one was non-binary. Four drove almost daily, three weekly, four monthly, and five rarely or never. Four had never used a VR headset, nine had tried one several times, and three were regular users; eleven had no experience of a steering-wheel simulator. Demographic information was collected in the post-study questionnaire. No completed session was excluded. No a priori power analysis was conducted; the sample size was constrained by the available recruitment window and the duration of the 21-round protocol for each participant. Participants were recruited through two channels: an open advertisement on a supervisor-maintained Discord server whose members were primarily students, and direct approaches by the author to acquaintances. Participation was unpaid, with no payment or voucher offered. This was stated in the participant information sheet before consent.

3.3 Experimental Apparatus

3.3.1 Simulator Hardware. Participants wore a Meta Quest 3S head-mounted display connected to the host computer by a wired link, with rendering performed on the host. Steering and pedal input were provided by a Logitech G25 steering wheel and pedal set. Inertial cues were presented by a Yaw VR motion platform, to which the simulator sent pose updates directly through the device’s network protocol (Figure 1 and Figure 2).



Figure 1: Experimental apparatus: Yaw3 three-degree-of-freedom motion platform, racing seat, Logitech force-feedback steering wheel, and three-pedal set. Participants sat wearing a Meta Quest 3S head-mounted display (the unoccupied apparatus is shown).



Figure 2: Side view of a participant during the experiment, with the face pixelated. The head-mounted display is connected by a wired Link cable, both hands are on the steering wheel, both feet are on the pedals, and the motion platform is in its operating position.

After the eighth participant, the pitch axis of the motion platform failed because the rubber on the friction wheel split and the pitch motor malfunctioned; it could not be repaired within the study period. P1–P8 therefore experienced inertial cues, whereas P9–P16 drove with the platform stationary. The simulator ran on a computer with an AMD Ryzen 5 9600X processor, 64 GB of DDR5-5600 memory, and an NVIDIA GeForce RTX 4070 Ti SUPER graphics card. The headset was connected directly through a wired Link connection, with a per-eye rendering resolution of 1832 x 1920

and a display refresh rate of 72 Hz. Table 1 gives the complete configuration. Frame rate was obtained from the Meta runtime’s own telemetry logs. During periods overlapping the experimental sessions, the mean frame rate was 71.99 fps, close to the 72 Hz display limit.

3.3.2 Virtual Environment. The simulator was built in Unity 6000.3.9f1. The virtual environment reproduces an area comprising Knightsbridge, Belgravia, Chelsea, and surrounding streets in central London. The visual streetscape was provided by Google Photorealistic 3D Tiles, streamed at runtime according to camera position through Cesium for Unity 1.23.3. The surface on which participants actually drove was a separately constructed road mesh based on OpenStreetMap geometry. The driveable network comprised 418 road segments with a total centreline length of 28.7 km and 564 junctions: 486 three-way junctions, 76 four-way junctions, and two five-way junctions.

Ambient traffic was simulated across the network with a commercial traffic system (Gley Traffic System v3.6.2), using priority control at junctions and traffic-light control at major junctions. Static pedestrians were placed on pavements. Participants drove a passenger car from a seated, first-person cockpit view. The interior rear-view mirror and both exterior mirrors rendered live rearward views onto the vehicle model’s own mirror surfaces.

3.3.3 Vehicle Dynamics. Vehicle control was configured for steering-wheel and pedal input. The vehicle used a torque curve, a maximum speed of 20 m/s (72 km/h), a driving force of 6500, a braking force of 9000 (simulator units), and a drag coefficient of 0.1. In practice, its terminal speed on level ground was approximately 52 km/h.

3.4 Eco-Driving Display and Design Space

The display comprised two types of spatial object. A leaf gauge, numerical eco-score, valence icon and advice text (jointly termed the framing element), speedometer, and acceleration icon were placed on a world-space canvas at the windscreen; vehicle markers were attached above surrounding traffic vehicles and faced the driver. A route-guidance arrow showing the next turn and its distance was always present and was not part of the design space.

The design space comprised seven continuous dimensions. Six independently set the size of one display element, and the seventh set overall display opacity. The optimiser operated on a normalised unit interval for each dimension, while the mapping from that interval to physical values was encapsulated in the display component.

The leaf gauge, numerical score, framing element, acceleration icon, and vehicle markers mapped from the unit interval to scale factors of 0–1.3. An element was not drawn when its physical scale fell below 0.6, corresponding to a normalised threshold of $0.6/1.3 \approx 0.462$. The speedometer mapped to 0.6–1.3 and was always retained. Opacity mapped to 0.10–1.0; the speedometer also retained its own minimum readable alpha (Table 2).

This parameterisation has two consequences worth stating explicitly. First, element presence is controlled by the same continuous variable as element size. This keeps the search space fully continuous, avoiding the additional complexity of mixed-variable Bayesian optimisation under a small evaluation budget, while still allowing

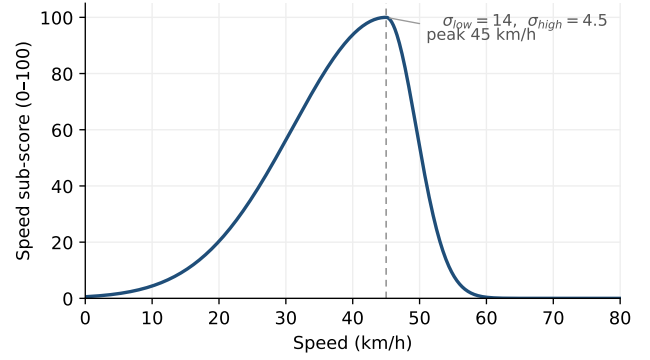


Figure 3: Speed sub-score curve: an asymmetric Gaussian peaking at 45 km/h, with $\sigma = 14$ km/h on the low-speed side and $\sigma = 4.5$ km/h on the high-speed side.

the optimiser to remove any element other than the speedometer by reducing its size to zero. Second, opacity controls overall salience while the colour coding remains at full intensity, so a dim display is uniformly dimmed as a whole.

Colour coding within the display represents the driver’s current eco-state: as the components of the eco-score change, the speedometer and acceleration icon transition between green, amber, and red.

3.5 Eco-Score and Energy Objective

The study used two distinct quantities that must not be conflated. The eco-score was visible to the driver during a round. The energy objective was returned to the optimiser after the round ended.

The eco-score construction followed Colley et al. [6]. The acceleration sub-score used the piecewise-linear form described in their Equation 2: the score was 100 when $|a| < 0.01$ m/s², zero above a cut-off, and decreased linearly between the two. The printed equation gives a slope of 33.9 and a cut-off of 2.95 m/s², which are not fully consistent with the adjacent statement that ‘a speed change of 100 km/h completed in 10 s should score zero’. The latter converts to 2.78 m/s² and corresponds to a slope of approximately 36.1. The present implementation followed that textual assumption and used a cut-off of 2.78 m/s² and a slope of 36.1.

In this study, the speed sub-score was reconstructed as an asymmetric Gaussian centred on the target speed, given in Equation 1.

$$V(v) = 100 \exp\left(-\frac{(v - v^*)^2}{2\sigma^2}\right), \quad \sigma = \begin{cases} \sigma_{\text{low}} & v \leq v^* \\ \sigma_{\text{high}} & v > v^* \end{cases} \quad (1)$$

where $v^* = 45$ km/h, $\sigma_{\text{low}} = 14$ km/h on the low-speed side, and $\sigma_{\text{high}} = 4.5$ km/h on the high-speed side.

The curve is shown in Figure 3. Its asymmetry was intended to impose only a gradual penalty below the target speed and a steeper penalty above it. Table 3 gives representative values.

The energy objective supplied to the optimiser was independent of the eco-score visible to the driver. It was an energy-intensity proxy based on simulated road load. At each time step, the model computed $F = ma + C_r mg + \frac{1}{2} \rho C_d A v^2$ and $P = Fv$. When $P \geq 0$, the energy increment was $P \Delta t / \eta_d$; when $P < 0$, it was $\eta_r P \Delta t$. The

Table 1: Simulator computer, software, virtual-reality and motion-platform configuration, and measured frame rate during the experiment. Hardware and runtime values were taken from Meta runtime performance logs, the memory-module layout was obtained from a system query, and the engine version was taken from the project settings.

Category	Item	Configuration
Host computer	Processor	AMD Ryzen 5 9600X (6 cores)
	Memory	2 x 32 GB DDR5-5600 (63,063 MB available)
	Motherboard	ASRock B650M Pro RS WiFi
	Graphics card	NVIDIA GeForce RTX 4070 Ti SUPER, 16 GB, driver 571.96
Software	Operating system	Windows 11 64-bit (build 26200.9168)
	Engine	Unity 6000.3.9f1
	VR runtime	Meta runtime 1.207.0 (OpenXR)
Virtual reality	Head-mounted display	Meta Quest 3S, direct display connection
	Refresh rate	72 Hz
	Connection	Wired Quest Link (USB3), encoding resolution 3680 x 1920
	Per-eye rendering resolution	1832x1920
Motion platform	Model and interface	Yaw3, 50 Hz pose stream, motion condition only
Measured frame rate	Within-session mean	71.99 fps
	Worst 60-second bin mean	70.42 fps
	5th-percentile bin mean	71.37 fps
	Compositor dropped frames	0.0587%
	ASW-synthesised frames	0%
	Telemetry coverage	35 telemetry bins, 73,245 frames, 16.96 minutes

Table 2: Seven-dimensional design space. Each dimension was normalised to the unit interval for the optimiser, while the physical mapping was encapsulated in the display component.

Dimension	Element controlled	Optimiser range	Physical mapping	Can vanish
size 1	Leaf gauge	0–1	x 1.3 (scale)	yes
size 2	Numeric score	0–1	x 1.3 (scale)	yes
size 3	Framing face + advice line	0–1	x 1.3 (scale)	yes
size 4	Speedometer	0–1	0.6 + 0.7x (scale)	no (research-design floor)
size 5	Acceleration pictogram	0–1	x 1.3 (scale)	yes
size 6	Vehicle markers	0–1	x 1.3 (scale)	yes
opacity	Whole display alpha	0–1	0.10 + 0.90x	no (floor 0.10)

Table 3: Speed sub-scores at representative speeds.

Speed (km/h)	25	30	32.2	40	45	47	50	53
Sub-score	36.0	56.3	65.8	93.8	100.0	90.6	53.9	20.6

implementation used $m = 1500$ kg, $C_d = 0.30$, $A = 2.2$ m², $C_r = 0.012$, $\rho = 1.225$ kg/m³, $\eta_d = 0.90$, $\eta_r = 0.60$, and $g = 9.81$ m/s², and the result was converted to kWh/100 km using the distance driven. The model included no explicit potential-energy term for road gradient and no separate collision-energy term; collisions entered only indirectly through the speed and acceleration trajectory. The parameters were not calibrated against a real vehicle. The measure is therefore interpreted only as an internally consistent simulated proxy. The optimisation bounds were 0–150 kWh/100 km; the upper bound was widened after truncation occurred in an early pilot.

3.6 Optimisation Setup

Optimisation used the differentiable expected hypervolume improvement (qEHVI) acquisition function. Daulton et al. [8] introduced this acquisition function by reformulating expected hypervolume improvement in differentiable form, allowing it to be solved with gradient methods and to propose multiple candidates at once.

It is implemented in BoTorch [1], a Bayesian optimisation framework built on PyTorch around Monte Carlo acquisition functions and automatic differentiation. Bayesian Optimization for Unity [13], an open-source Unity toolkit, bridged the simulator and optimiser by running BoTorch as a separate process and handling parameter delivery and objective return. The optimiser ran as an independent process on the host computer and communicated with the simulator through a local socket.

The optimisation problem had seven design dimensions and five objectives (Table 4). The first 15 rounds were Sobol samples from the unit hypercube, followed by five qEHVI rounds with a batch size of one. Round 21 repeated a candidate already observed during the first 20 rounds and did not propose a new design. The selector first chose from the observed Pareto set by minimising normalised distance to the ideal point; ties were resolved in sequence by maximising the minimum margin, minimising aggression, and choosing the earlier round. Of the 16 recommended configurations, eight originated in the sampling phase and eight in the optimisation phase.

The acquisition function used 512 Monte Carlo samples, 1,024 raw samples, and ten restarts. The 15 initial points followed the 'twice the dimensionality plus one' heuristic. Compared with the five sampling and ten optimisation rounds used by OptiCarVis, this study allocated more of its budget to covering the seven-dimensional space and consequently retained only five model-driven iterations.

One detail of the initial design is worth recording. Because elements were hidden when their sizes approached zero, Sobol sampling could in principle produce a configuration in which all optimiser-controlled elements were invisible. Such samples were detected and repaired before presentation, so no participant experienced a round with no eco-driving feedback at all.

3.7 Measures

At the end of each round, participants answered five items about that round's display using 21-position discrete sliders labelled from 0 to 20 and marked with tick marks (Table 5).

The first two items were averaged to form the task-load objective, which the optimiser minimised; the final three were each maximised as independent objectives. Their separation was deliberate: the same design change could move informativeness, pleasantness, and legibility in opposite directions, and averaging would conceal the conflict that the optimiser was intended to negotiate. The load items were worded so that higher values were undesirable, whereas the acceptance items used higher values for desirable responses, avoiding a built-in positive correlation between the two groups. Item sources are shown in Table 5. Usefulness was operationalised as informativeness because fuel consumption could not be perceived directly in the simulator. Single-item measures were used because every design had to be rated after each of 21 rounds.

The questionnaire imposed one data-quality constraint. Each slider opened at the midpoint, and the submit control remained disabled until every slider had been moved at least once. A missing response and an intentional midpoint rating would otherwise be passed to the optimiser as the same value; the movement requirement distinguished them. Slider positions did not carry over from the preceding round, so the previous rating did not anchor the next one. Objective driving data, including speed, acceleration, components of the eco-score, and energy use, were recorded throughout every round.

After the experiment, participants completed a post-study questionnaire in Qualtrics, UCL's survey platform. It collected demographics and prior experience: age, gender, years since obtaining a driving licence, driving frequency, and experience with VR headsets and steering-wheel simulators. It further contained five agreement items on five-point scales about the final display design, one five-point item on dizziness, nausea, or eye strain, and one item on the experience of seat movement. Two open questions asked which display element they liked and disliked most and whether the final design was their ideal design.

3.8 Driving Task and Routes

Each round consisted of one drive along one of four specified urban routes; their positions and starting points in the road network are shown in Figure 4. R1 was 215 m long and R2 was 165 m long, and both were straight. From R2 onwards, stationary pedestrians

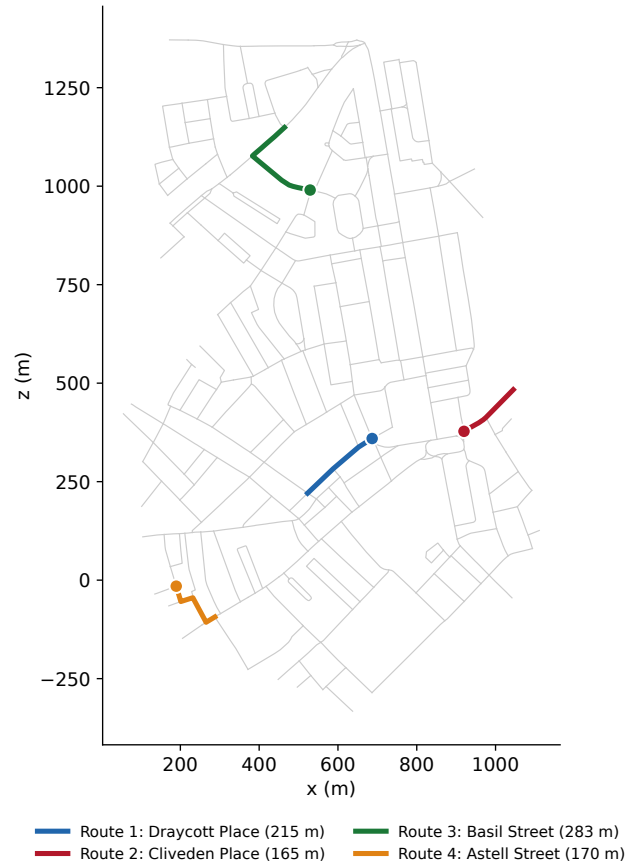


Figure 4: Driveable road network (grey), the four experimental routes, and their starting points.

were added to the pavements. R3 was 283 m long and included one turn greater than 30°; R4 was 170 m long and included three turns. A windscreen guidance arrow indicated the next turn and the remaining distance. Participants drove among ambient traffic and traffic lights, and the round ended when the route was completed. A round was terminated early if its duration exceeded a route-length-scaled limit, preventing a stuck vehicle from halting the entire session.

3.9 Experimental Procedure

After an introduction and written informed consent, participants put on the head-mounted display and sat at the steering wheel. They were first given time to become familiar with the vehicle controls, steering wheel, pedals, and motion platform. They then completed 21 optimisation rounds, with the five-item questionnaire presented in the virtual environment immediately after each round. The experiment concluded with the post-study questionnaire and a brief debriefing conversation.

The median duration of the in-simulator portion (driving and per-round questionnaires) was 24.2 minutes (IQR 18.2–32.1, range 16.5–38.0). Under a separate component-based accounting of the logs, mean driving time was approximately 14.9 minutes, mean

Table 4: The five optimisation objectives, their bounds, directions, and sources.

Objective	Bounds	Direction	Source
Simulated energy proxy (kWh/100 km)	0–150	minimise	road-load integral over the driven round
Task load	0–100	minimise	mean of survey items 1–2
Informativeness	0–100	maximise	survey item 3
Pleasantness	0–100	maximise	survey item 4
Legibility	0–100	maximise	survey item 5

Table 5: Five items in the per-round questionnaire. The panel title was “Rate THIS round’s display”, with scale anchors from ‘0 = not at all’ to ‘20 = very much’. Each item used a 21-position integer slider that opened at the midpoint of 10. Items 1 and 2 were averaged to form the task-load objective, while Items 3–5 each formed a separate acceptance objective. The four subjective objectives were linearly mapped to 0–100 before being passed to the optimiser. Single-item measures were used to keep the per-round questionnaire within a duration sustainable for the optimisation loop.

#	Item (wording shown to participants in VR)	Range	Construct	Optimisation objective	Direction	Source
1	It was mentally demanding	0–20	Mental demand	Task load	Minimise	NASA-TLX mental-demand subscale
2	It distracted me from the road	0–20	Attentional interference	Task load	Minimise	Author-developed; detects interference with the primary task
3	It kept me informed about my driving	0–20	Usefulness, operationalised as informativeness	Informativeness	Maximise	Based on the van der Laan acceptance scale
4	It was pleasant to use	0–20	Satisfaction	Pleasantness	Maximise	Based on the van der Laan acceptance scale
5	I could read it at a glance	0–20	Glance legibility	Legibility	Maximise	Author-developed for glance-based use of a head-up display

questionnaire time approximately 10.4 minutes, and cumulative mean optimisation-computation time approximately 6.7 minutes.

3.10 Pilot Calibration

The author completed one 36-round manual pilot to verify the workflow and set approximate bounds for the energy objective. The minimum pilot observation was 44.8 kWh/100 km. The original upper bound was 100, and four of the 36 rounds were truncated to 100.000 before being written to disk, so their true values cannot be recovered; the upper bound was therefore widened to 150 for the formal study. This boundary decision was an engineering adjustment made after seeing the pilot results, not a preregistered validation. More importantly, the pilot preceded finalisation of the vehicle dynamics and motion platform, so its system version differed from that used in the formal experiment. Its role was limited to identifying truncation risk and confirming that the workflow could run.

3.11 Ethics and Data Protection

The study was approved by the UCL Research Ethics Committee under project number 1165. Before the experiment, participants received a participant information sheet and signed a written consent form; the task instructions and the content of both documents are reproduced in Appendix G. The study was conducted in the laboratory. The information sheet explained that virtual reality could cause nausea, eye strain, and disorientation and specified

responses including stopping the study, removing the headset, and resting if discomfort occurred. Participation was voluntary and unpaid; participants could withdraw without giving a reason and could specify how data already collected should be handled. The participant shown in Figure 2 separately consented to research photography and use of the photograph in the thesis; the face has been pixelated in this draft.

Exclusion criteria covered a history of binocular-vision abnormalities, mental illness, heart disease, implanted medical devices, seizures, photosensitivity, or severe vertigo. These criteria were stated in the information sheet and confirmed on the consent form.

Research data were stored under pseudonymous identifiers, and the thesis uses labels such as P1 and P2. The existing materials are not fully consistent in their descriptions of open access: the participant information sheet states that anonymised data may be made public, whereas the consent form states that pseudonymised data may be used by authenticated researchers in future research. Per-round data and high-dimensional driving trajectories are therefore not described as already public before the principal investigator and ethics review have confirmed the permitted scope of release. Any subsequent release must first undergo a re-identification-risk assessment and comply with the finally approved data-governance arrangements.

3.12 Analysis

The unit of analysis was the participant, with rounds treated as repeated measures. Every inferential comparison was therefore calculated using either one value per participant or one value per participant per phase; the 336 mutually dependent rounds were not treated as independent observations. Phase had three levels: sampling (Rounds 1–15), optimisation (Rounds 16–20), and final design (Round 21), with the final design being a repeat drive of a design evaluated in an earlier round. Analysis was conducted in R 4.6.1. ARTool provided the aligned rank transform analysis of variance used for omnibus phase tests. moocore and emoa independently computed Pareto fronts and hypervolume and were used to cross-check one another. lme4 and lmerTest fitted linear mixed models and supplied significance tests for random-slope likelihood-ratio comparisons. emmeans computed estimated marginal means; effectsize computed effect sizes and their intervals; rmcorr computed within-participant repeated-measures correlations; and colleyRstats formatted the results consistently. Package versions were ARTool 0.11.2, moocore 0.3.2, emoa 0.5-3, lme4 2.0-6, lmerTest 3.2-1, emmeans 2.0.4, effectsize 1.0.3, rmcorr 0.7.0, and colleyRstats 0.1.6. Continuous variables were summarised with means and standard deviations. Non-parametric or rank-based tests were used throughout because the participant-level sample was small (16, or eight per group after grouping), a size at which normality could not be established. Exact *p* values are reported to three decimal places.

RQ1 was examined from four perspectives because no single summary is sufficient for a five-objective problem. First, an exact two-sided Wilcoxon signed-rank test compared hypervolume after the final optimisation round with hypervolume at the end of sampling, reporting *V*, the paired rank-biserial correlation, and paired-samples Cohen's *d*. Second, because hypervolume is cumulative and can increase simply by drawing five additional designs, the gain over the five optimisation rounds was compared within each participant with the gain over the immediately preceding five sampling rounds. Cumulative hypervolume was recomputed after each evaluation in the normalised minimisation space, with reference point (1, 1, 1, 1, 1). Third, each objective was aggregated to one value per participant per phase, an aligned rank transform (ART) analysis of variance tested the omnibus phase effect, and a Friedman test provided a rank-based cross-check. Both were omnibus tests across three levels; no pairwise post-hoc comparisons were calculated, so they were reported only as evidence that phase mattered. Fourth, per-round Pareto hit rates were compared between the two phases. A round was Pareto-optimal when no other round for the same participant was no worse on all five objectives and strictly better on at least one. Fronts were calculated independently with moocore and emoa, and both implementations were checked against the optimiser's recorded flag for every round.

The final recommended design was compared with a route-matched sampling baseline: that participant's sampling round on the route driven in Round 21. Matching controlled only route identity; it did not control phase order, practice, traffic state, rating drift, or within-route differences in driving. The five comparisons were Holm-corrected across the five objectives within the same family, with both raw and adjusted *p* values reported.

RQ2 was examined from three perspectives. First, between-participant dispersion was reported for each dimension and compared with the mean within-participant standard deviation over the five optimisation rounds. Second, a one-sided label-permutation test with 5,000 permutations examined whether the pooled Pareto designs were closer to others carrying the same participant label than under random relabelling. The test preserved the pooled sample size but could not separate participant preference from different initial Sobol points and the search paths they generated; the result is therefore described as participant-associated clustering. Random-slope likelihood-ratio tests then examined whether parameter-objective associations varied by participant. The transfer analysis used ridge-regression response surfaces, but was interpreted exploratorily only after leave-one-round-out predictive validity had been reported.

RQ3 was summarised descriptively using experimental logs and the post-study questionnaire. The motion and no-motion groups were compared using exact two-sided Mann-Whitney tests on session-level summaries and were not pooled before this check. All analyses were run by scripts from the raw logs through to every reported value.

The sample size was not determined by an a priori power analysis, but by the available recruitment window and the duration of the 21-round protocol for each participant; a sensitivity analysis is therefore reported. With two-sided $\alpha = .05$ and power of .80, the minimum detectable effect for a within-participant paired comparison ($N = 16$) was $dz = 0.75$, or approximately 0.77 when using the signed-rank test. For comparisons between the motion and no-motion groups (eight participants per group), the minimum detectable effect was $d = 1.51$, or approximately 1.54 with a Mann-Whitney test. The design was therefore sensitive to at least moderate within-participant effects, but only to very large between-group effects.

4 System Implementation

This section describes the implementation of the simulator and optimisation pipeline because engineering decisions directly delimit what could be measured. The public repository listed in the Open Science section currently provides self-authored Unity/Python source code and documentation; it should not be understood as already containing the complete data, analysis snapshot, or third-party assets.

4.1 Environment Construction: Three-Dimensional City Tiles and Driveable Roads

The environment combined two geometry sources with different roles. Google Photorealistic 3D Tiles were streamed into the scene through Cesium for Unity 1.23.3 and provided the neighbourhood's visual identity, including red-brick façades, rooflines, and street furniture. These tiles are photogrammetrically reconstructed surfaces: the mesh is uneven at wheel scale, parked vehicles and vegetation are baked into it, and the road areas are neither flat nor provided with clean collision geometry. The tiles therefore served visual presentation only. The driveable surface was generated separately by CityGen3D from OpenStreetMap data and provided geometrically



Figure 5: Overview of the virtual environment, looking directly down over the Knightsbridge area. Google Photorealistic 3D Tiles streamed through Cesium provide the neighbourhood appearance.



Figure 6: The same viewpoint with the tiles disabled, leaving only the driveable road surface, kerbs, and pavements generated by CityGen3D from OpenStreetMap data. Vehicle physics operated entirely on this surface.

correct carriageways, kerb lines, and pavements. Vehicle physics operated entirely on this surface.

Coexistence of the two sources required alignment and clipping. The tile set was georeferenced and rigidly aligned with the OpenStreetMap road network so that façades stood on the correct side of each street. The photogrammetric surface was then clipped along a corridor above the carriageway with a height threshold, allowing tree canopies and buildings that extended over the road to remain (Figure 5, Figure 6, and Figure 7).

4.2 Windscreen Display

The HUD comprised a world-space canvas fixed at the windscreen and separate world-space markers attached above surrounding vehicles. The static canvas carried the leaf gauge, numerical eco-score, framing element, speed group, and acceleration icon. Vehicle markers used SpriteRenderer to face the driver and were shown or hidden according to the windscreen viewing aperture, occlusion, and distance threshold. The seven design parameters controlled the scale of the six element types and the overall opacity, respectively



Figure 7: Forward view while driving.

(Figure 8 and Figure 9). Three in-vehicle mirrors were supplied with live images from rear-facing cameras through render textures.

4.3 Motion-Platform Integration

The motion platform provided physical acceleration cues during driving. The system applied tilt coordination to low-pass-filtered longitudinal and lateral acceleration, with pitch limited to $\pm 12^\circ$, roll to $\pm 10^\circ$, rate limited to $20^\circ/\text{s}$, and washout towards the neutral position at $4^\circ/\text{s}$; yaw followed vehicle heading. The 50 Hz pose stream means that the simulator sent target pitch, roll, and yaw angles to the platform controller over UDP 50 times per second, allowing the controller to track the vehicle state continuously. TCP was used only to establish the connection and write device-side travel limits; the limits were written before motion began and were thereafter enforced by the device. Figure 10 shows the data flow for discovery, connection, limit configuration, and continuous pose updates. The link was first tested using the manufacturer's desktop simulator and then validated on the physical platform.

4.4 Vehicle Physics on a Height-Map Road Surface

The driveable surface was a terrain height map, so junctions and road crowns had physical gradients. The vehicle-body collider was narrowed to the visible body and raised above the ground and was responsible only for collisions with walls and other vehicles. Four low-friction spherical colliders at the model's actual wheel positions supported the vehicle and could roll over gradients, dropped kerbs, and terrain-tile seams.

Energy use was integrated along the trajectory using the road-load model described in subsection 3.5 and converted to kWh/100 km. It was independent of the eco-score visible to the driver and was used only as a simulated energy-intensity proxy.

4.5 Vehicle Control and Steering-Wheel Input

The steering wheel and pedals were read through the operating system's native joystick interface rather than the game engine's input layer. Before each session, an initialisation component placed the device in native mode: 900° of steering rotation, with steering and the two pedals each occupying an independent axis. Steering was

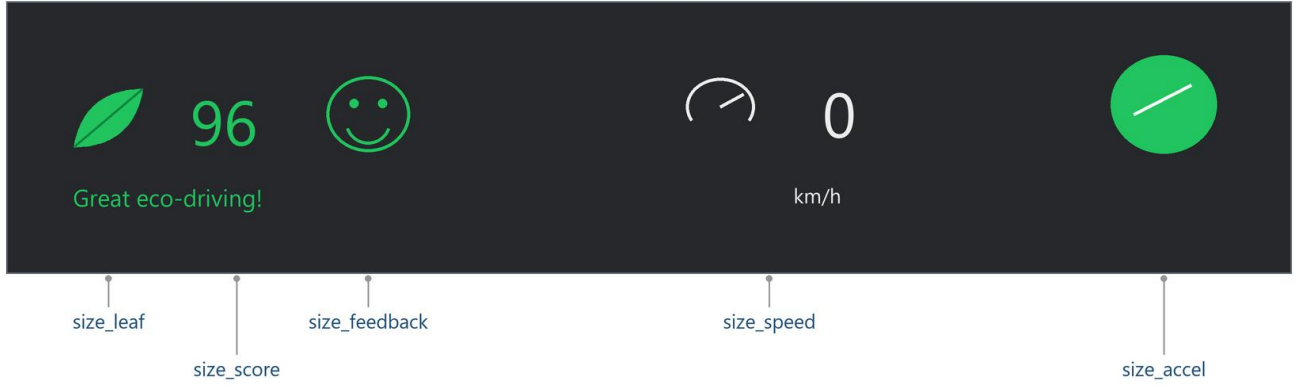


Figure 8: Six adjustable element types in the eco-driving display. The illustrated windscreen canvas shows five static channels: the leaf gauge, numerical eco-score, framing element (valence face and advice text), speed group, and acceleration icon. The sixth type comprises world-space markers drawn above surrounding vehicles at runtime and is not part of this static canvas.



Figure 9: Effects of the size and opacity mechanisms, captured at the same location, under the same lighting, with the vehicle stationary. Top left: elements at a large size. Top right: the same elements at a smaller size. Bottom left: optimiser-controlled elements below the hiding threshold, leaving only the speedometer and the route-guidance arrow on the instrument cluster, neither of which can be removed by the optimiser. Bottom right: the entire canvas at its minimum opacity, with all elements uniformly dimmed while colour coding remains at full intensity; the speed reading remains legible because it has its own readability floor.

a signed axis, and a dead zone was applied in the raw input range before mapping to the steering range, preventing slight off-centring of the wheel from producing steering input. The shift paddles acted as a two-speed gearbox, with the right paddle selecting forward and the left selecting reverse. Analogue and keyboard inputs were combined by taking the greater magnitude, allowing the experiment to continue if the steering wheel failed.

4.6 Ambient Traffic and Its Behaviour

Ambient traffic ran on a commercial traffic system, over which the complete neighbourhood road network was constructed programmatically from an exported road graph. Each junction-to-junction segment generated an open road; each junction used priority control and generated stopping waypoints for every incoming lane; lane-connection curves across junctions were matched by position;

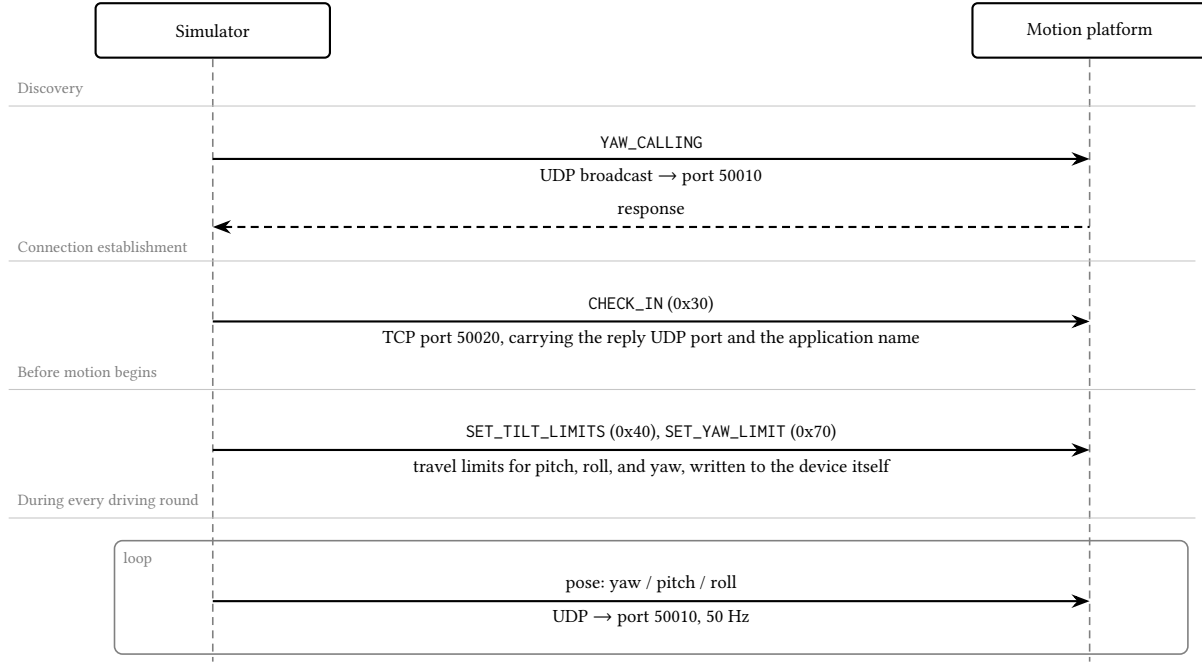


Figure 10: Connection and data flow between the simulator and motion platform. Discovery and the pose stream use UDP; connection establishment and travel-limit writing use TCP. Travel limits are written to the device before motion begins and are subsequently enforced by the device itself.

and major junctions along the study routes were given signal control and stop lines. Three additional behaviours were required for these narrow, two-way streets.

First, speed variation: each traffic vehicle carried an individual cruising factor sampled from a range below the road’s configured speed and resampled at staggered intervals, so vehicles travelled at different, slowly changing speeds without exceeding the road setting.

Second, overtaking was deliberately disabled. On these narrow two-way streets, the native overtaking behaviour changed to a waypoint in the opposing lane, effectively driving against traffic. It was replaced by dispersal of queues that inevitably formed behind slow drivers: vehicles that had remained off-camera immediately behind the player and moving slowly for longer than a threshold were silently removed, and the density system respawned them elsewhere after an overlap check.

Third, head-on deadlocks were resolved. The native behaviour could leave two vehicles approaching one another permanently stopped on a narrow street. All vehicle pairs were therefore scanned every 0.3 s, and a pair was classified as an imminent encounter when its heading dot product was below -0.6 , separation below 28 m, distance still closing, and extrapolated lateral separation below 2.6 m. Both vehicles were then pulled towards their respective lane edges and slowed to pass.

Finally, an encounter-guarantee mechanism ensured that manipulation of vehicle markers was meaningful in every round. Marker visibility was an optimised dimension, but three systems reduced the number of vehicles that could be marked: a 30 m bubble around



Figure 11: First-person view while driving (Draycott Place, Route R1): an ambient traffic vehicle ahead, a signal-controlled junction, the speed reading on the windscreen (19 km/h), route guidance on the instrument cluster (“20 m to finish”), and the live rearward view in the interior mirror.

the player was cleared at the beginning of each round, the queue disperser removed followers, and the encounter resolver cleared deadlocked pairs. The mechanism therefore guaranteed that at least one ambient vehicle appeared in view during every round (Figure 11).



Figure 12: Questionnaire presented in the virtual environment after each driving round. Participants answered five items with integer sliders from 0 to 20: mental demand, distraction from the road, informativeness about driving, pleasantness of use, and at-a-glance legibility. The endpoints were ‘0 = not at all’ and ‘20 = very much’. Each slider started at the midpoint of 10 and had to be moved at least once before submission.

4.7 Experimental Loop

The complete experiment was driven by a round controller. For each round, it applied the candidate design to the display, placed the vehicle at the route starting point, ran the drive, presented the questionnaire in the virtual environment, returned the five objective values to the optimiser, and received the next candidate. The optimiser ran as an independent process and communicated through a local socket. During testing, a development-only automated-driving surrogate replaced the participant. It was used solely as a development aid and generated no research data. Each round wrote the seven design parameters, five objective values, current phase, and Pareto membership of the observation to a per-participant log.

4.8 Implementation of the Questionnaire in Virtual Reality

At the end of each round, the questionnaire appeared in front of the vehicle as a world-space canvas, suspended above the bonnet in the driver’s forward view, with the windscreen display still visible behind it. The panel contained five integer sliders from 0 to 20, with 21 tick marks along each track to align with the integer positions, a live numerical readout on the right, and a submit button below. The button remained disabled until all five sliders had been moved. Responses were made using a ray. The ray preferentially used the aiming pose from Quest hand tracking (the right hand by default); when no hand-aim pose was available, it fell back to the pointing pose of a Touch controller. Confirmation was an index-finger pinch under hand tracking, with hysteresis to prevent unintended activation from jitter, and the trigger under controller input. The intersection between the ray and canvas was converted to screen coordinates and passed to Unity’s graphic raycaster, thereby reusing the standard pointer-enter, press, drag, and click events and allowing the sliders to be dragged directly. The ray was visualised as a thin blue line so that participants always knew where they were pointing (Figure 12).

4.9 Route Generation

The driveable road network was exported as a graph whose nodes were junctions and whose edges were road segments; each edge stored length and a lane half-width derived from road class. The generator performed a depth-first search over the graph, enumerating paths that did not revisit a road segment and imposing an upper bound on the number of edges to control search size. Candidates had to satisfy three conditions simultaneously: total length within the range suitable for a single driving round, cumulative turning angle above a lower bound to exclude entirely straight paths, and avoidance of a predefined excluded area. Eligible candidates were ranked by score and then spatially deduplicated: only one was retained within each 250 m grid cell, and its nearest distance to any already selected route had to be at least 30 m, distributing selected routes across the neighbourhood. One route encircling a block was generated instead through a cycle search, and a point-in-polygon test confirmed that the cycle enclosed the target block.

Routes were taken directly from the road-network graph, so their polylines lay on the road centreline and each point carried the lane half-width at that location. The guidance arrow, development-only automated-driving surrogate, and route-completion test all read the same polyline. Candidates were drawn as an overlay above the neighbourhood for visual inspection; the overlay was rendered only in the editor.

5 Results

This section reports, in research-question order, the multi-objective indicators across the fixed-order phases, the recommended configurations and same-route references, participant-associated design differentiation, and feasibility and device-condition sensitivity analyses. Every value was checked against the local analysis output; the present and planned release status of reproducibility materials is described in the Open Science section.

5.1 Dataset

All 16 participants (P1–P16) completed all 21 rounds, giving 336 observations: 15 sampling rounds, five optimisation rounds, and one final-design round per participant. P1–P8 completed the protocol with the motion platform active. After failure of the platform’s pitch axis, P9–P16 completed it with the seat stationary, producing two equally sized groups of eight. All completed sessions entered the analysis; none was excluded.

5.2 Within-Session Optimisation Progress (RQ1)

5.2.1 Hypervolume Progress. Between the end of sampling at Round 15 and the end of optimisation at Round 20, cumulative hypervolume increased for 15 of the 16 participants and remained unchanged for one; the median relative gain was 10.9%. The paired comparison was $V = 135$, $p < .001$ (exact $p = .00006$), $r = .71$, Cohen’s $d = 0.86$, 95% CI [0.28, 1.43]. Because hypervolume is constructed to increase monotonically when new non-dominated points are added, and because phase order was fixed, this result describes the change in the cumulative front after five additional evaluations.

As a budget-matched internal sensitivity comparison, the hypervolume gain across the five optimisation rounds exceeded that

across the immediately preceding five sampling rounds: $V = 107$, $p = .044$, $r = .36$, $d = 0.58$, 95% CI [0.04, 1.10]. The two five-round windows nevertheless occurred at different times in the session and were based on different search mechanisms and paths, so this remains an internal comparison under a fixed ordering (Figure 13).

5.2.2 Comparison of the Sampling and Optimisation Phases. The five objectives formed a single outcome family. Raw p values are retained in the table for auditability, and the ART and Friedman results were separately Holm-corrected across the five objectives. After correction, the omnibus phase associations remained significant for the four subjective objectives but not for the simulated energy-intensity proxy. These omnibus tests do not identify which two phases differ and do not remove the fixed-order confound.

The omnibus phase test for task load gave ART $F(2, 30) = 10.34$, raw $p < .001$, partial $\eta^2 = .41$; the Friedman result was $\chi^2(2) = 11.38$, raw $p = .003$, Holm $p = .006$. Raw ART p values for informativeness, pleasantness, and legibility were all $< .001$ and remained $< .005$ after Holm correction; their Friedman Holm p values were $< .005$, $.006$, and $< .005$, respectively. The simulated energy-intensity proxy showed no omnibus phase association: ART $F(2, 30) = 2.37$, $p = .111$; Friedman $\chi^2(2) = 2.00$, Holm $p = .368$ (Table 6).

Because the three phases occurred in a fixed temporal order, round number, practice, fatigue, rating strategy, and model-driven selection all covaried with phase. Figure 14 and Figure 15 therefore show round trajectories descriptively and do not treat the trend lines as causal effects of phase.

Figure 15 presents the simulated energy-intensity proxy using the same conventions. It shows substantial round-to-round fluctuation, and the omnibus phase test was non-significant; the linear trend is descriptive only and does not provide evidence that energy use changed with optimisation.

5.2.3 Pareto Hit Rate. To avoid averaging five competing objectives into a single score, I independently recomputed each participant's Pareto front across the first 20 rounds in five dimensions: the simulated energy proxy and four rating objectives. moocore and emoa agreed across all 320 rounds and matched the log flags; 93 rounds were Pareto-optimal within their participant (Table 7). This hit rate depends on all five observed objectives, including the four participant ratings.

Calculated per round, the Pareto hit rate was 0.24 in the sampling phase and 0.44 in the optimisation phase; the paired comparison gave $V = 111$, $p = .024$, $r = .40$. The difference is in the direction expected from model-driven selection, but the two phases were fixed in sequence and the front was defined retrospectively using all 20 rounds. It is therefore described as a fixed-order phase association, not interpreted as a causal doubling of the hit rate by the optimiser (Figure 16).

5.2.4 Final Recommended Design. Round 21 repeated a recommended configuration observed during the first 20 rounds. The selector chose from the observed Pareto candidates using normalised distance to the ideal point and predefined tie-breaking rules; eight of the 16 recommended configurations came from sampling and eight from optimisation. Relative to the same-route sampling baseline, the recommended configuration improved task load, informativeness, pleasantness, and legibility by 15.8, 19.8, 16.6, and 18.2 points,

respectively, and all remained significant after Holm correction within the five-objective family. The difference in the simulated energy proxy was -0.4 kWh/100 km, $V = 62$, $p = .782$, Holm $p = .782$. Matching controlled route identity but not order, practice, traffic, or repeated-measurement conditions (Table 8).

The only comparison in which the selected design was inferior was against that participant's own best sampling round: the latter used 7.3 kWh/100 km less energy and was also slightly better on the subjective objectives. This is expected because the optimiser selected a compromise from across the front, whereas the 'best of 15 rounds' baseline is itself a maximum selected from a larger sample.

5.2.5 Measurement Properties of the Energy Objective. Energy was the only objective that did not move, so its measurement properties were characterised directly. Three quantities are reported as measurement results.

First, a post-hoc sensitivity analysis of the configured bounds. The energy proxy was declared to the optimiser as [0, 150] kWh/100 km, whereas observed values ranged from 16.0 to 64.6, occupying only 32.4% of that interval, with a normalised standard deviation of 0.060. When the bounds were changed post hoc to [15, 65], the internal five-round-window comparison changed from $V = 107$, $p = .044$, $d = 0.58$ to $V = 117$, $p = .009$, $d = 0.80$ (Table 9). Because the new bounds were determined from the same data, this result shows only that the conclusion was sensitive to scaling; it is not independent evidence of a validated performance improvement or of how alternative bounds should be selected.

Second, cross-round retest differences for the same configuration. Round 21 repeated a previously presented configuration, but retesting also changed the point in the session and could change route state, traffic interaction, and driving familiarity. Mean energy-proxy values in the source and retest rounds were 29.318 and 34.382 kWh/100 km, respectively, and the paired difference was significant (source minus retest = -5.064 , $p = .00336$). This difference therefore cannot be called pure 'measurement noise'. It combines variation from session timing, route or traffic, driving behaviour, and error in the proxy model, and it also shows that the energy proxy from one short-route round was unstable.

Third, variance-association models showed that, in the fitted within-participant models, the proportion of harsh acceleration and mean absolute acceleration were associated with the energy proxy, whereas mean speed had only a weak association. Driving-behaviour covariates corresponded to a model increment of approximately 59.1%, and the seven display parameters to approximately 5.1%. In a model restricted to sampling rounds, the incremental test of display parameters against the energy residual was $\chi^2(7) = 13.6$, $p = .059$. These percentages are associative or model-attributed shares under a particular specification, not causal proportions by which the variables 'explain' real energy use, and they do not establish either that the display is unrelated to energy or the opposite.

5.3 Differentiation of Personalised Designs (RQ2)

5.3.1 Parameter Values of the Recommended Configurations. Across the recommended configurations in Round 21, six of the seven dimensions covered almost the full normalised range. The

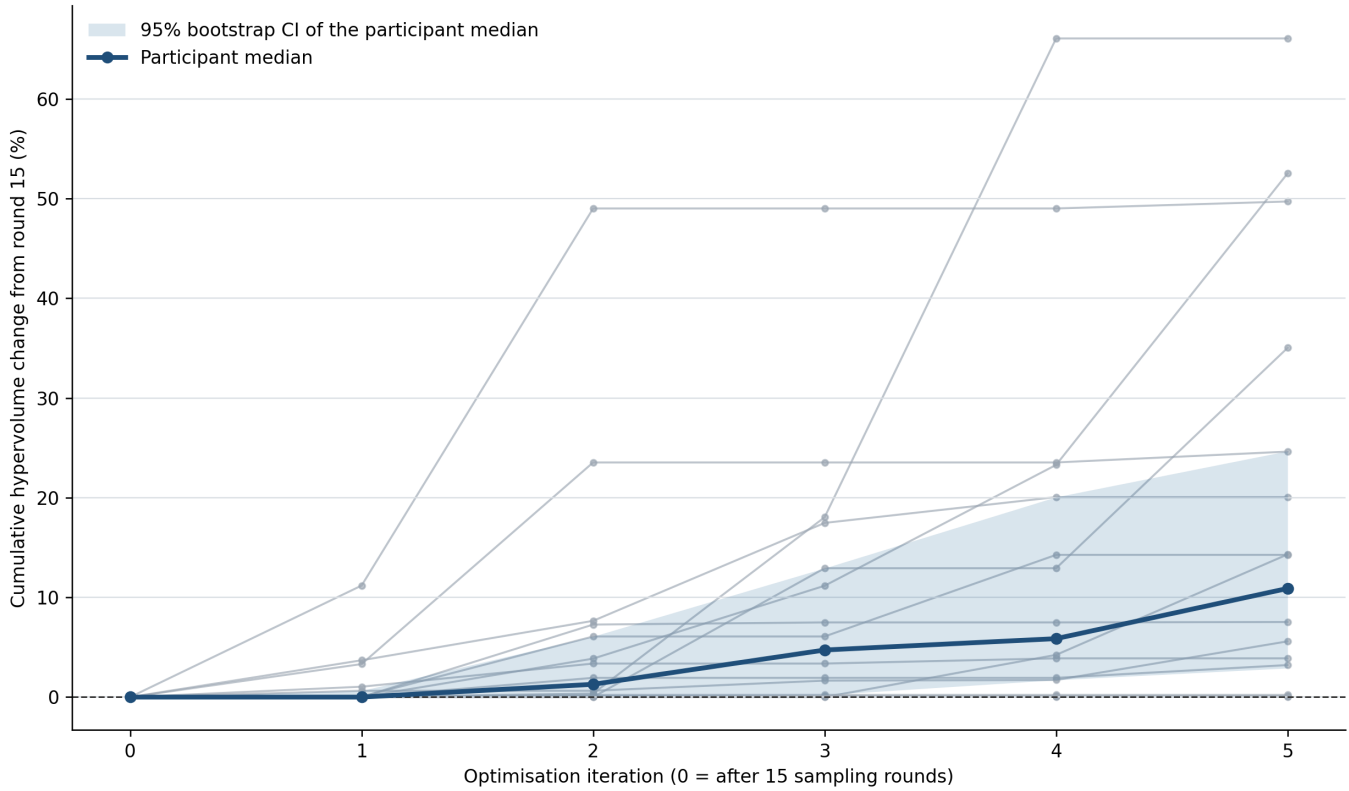


Figure 13: Percentage change in participants' cumulative hypervolume relative to its value at the end of sampling in Round 15. Thin lines represent individual participants; the thick line is the median at each round, and the shaded area is a participant-level bootstrap 95% confidence interval. The dashed line marks the five model-driven optimisation rounds. An increase in this cumulative metric should not, by itself, be interpreted as a causal optimisation effect.

Table 6: Omnibus associations between phase and the five objectives ($N = 16$; one value per participant per phase; phases = sampling, optimisation, and final design). The table reports raw ART and Friedman p values; the text separately reports the conclusions after Holm correction across five objectives for each test. The tests do not provide pairwise phase comparisons, and the fixed phase order also limits causal interpretation.

Objective	ART $F(2, 30)$	p	partial eta squared	Friedman chi-sq(2)	p
Simulated energy proxy (kWh/100 km)	2.37	.111	.14	2.00	.368
Task load	10.34	< .001	.41	11.38	.003
Informativeness	22.44	< .001	.60	16.62	< .001
Pleasantness	15.98	< .001	.52	12.88	.002
Legibility	22.84	< .001	.60	13.88	< .001

numerical-score dimension was narrower ($M = 0.37$, $SD = 0.21$, maximum = 0.77). These values describe configurations selected from observed candidates; they should not be called converged or stable individual optima, because each participant provided only 20 modelling evaluations and one retest (Figure 17, Table 10).

5.3.2 Which Elements Were Retained. In the final designs, the leaf gauge was retained for 11 of the 16 participants, while the acceleration icon and vehicle markers were each retained for eight, the framing element for seven, and the numerical score for five. No optimiser-controlled element was retained by everyone, and

none was removed by everyone. The speedometer could not be suppressed by design and is excluded from these counts (Figure 18).

5.3.3 Are Between-Participant Differences Larger Than Variation Within the Optimisation Rounds? For every parameter, the between-participant standard deviation of the recommended values exceeded the mean within-participant standard deviation across the five optimisation rounds, with ratios of 1.34–1.89. This indicates participant-associated differentiation in the selected configurations, but the comparison also contains variation caused by different initial samples, search paths, and the limited budget; it cannot by itself establish stable individual preferences.

Table 7: Pareto-optimal rounds by participant. A round was Pareto-optimal when no other round for the same participant was no worse on all five objectives and strictly better on at least one. Fronts were calculated independently with moocore and emoa, which agreed on all 320 rounds.

P	Pareto rounds (of 20)	In sampling (of 15)	In optimisation (of 5)	Rate sampling	Rate optim.
P01	4	4	0	0.27	0.00
P02	9	6	3	0.40	0.60
P03	8	7	1	0.47	0.20
P04	5	2	3	0.13	0.60
P05	7	6	1	0.40	0.20
P06	3	0	3	0.00	0.60
P07	7	4	3	0.27	0.60
P08	9	7	2	0.47	0.40
P09	9	5	4	0.33	0.80
P10	2	1	1	0.07	0.20
P11	2	1	1	0.07	0.20
P12	5	4	1	0.27	0.20
P13	7	4	3	0.27	0.60
P14	8	5	3	0.33	0.60
P15	5	1	4	0.07	0.80
P16	3	1	2	0.07	0.40
All	93	58	35	0.24	0.44

Table 8: Comparison of the recommended configuration in Round 21 with two references ($N = 16$; exact two-sided Wilcoxon signed-rank tests; d is paired-samples Cohen’s d ; Holm correction was applied across five objectives within each reference family). The same-route sampling baseline matched route identity only. The best sampling round takes the best value from the 15 rounds separately for each objective and therefore does not represent one common design.

Objective	Dir.	Final M	Matched-route M	Delta	p	Holm p	d	vs best sampling round
Simulated energy proxy (kWh/100 km)	min	34.4	34.0	-0.4	.782	.782	-0.08	worse by 7.3, Holm $p = .001$
Task load	min	20.3	36.1	15.8	.009	.018	0.84	n.s. (Holm $p = 1.00$)
Informativeness	max	85.3	65.5	19.8	< .001	< .001	1.15	n.s. (Holm $p = 1.00$)
Pleasantness	max	83.8	67.1	16.6	.004	.013	0.85	n.s. (Holm $p = 1.00$)
Legibility	max	84.4	66.2	18.2	< .001	.003	1.20	n.s. (Holm $p = 1.00$)

Table 9: Occupancy of the configured range by each objective and the resulting leverage in the normalised space used by the optimiser.

Objective	Configured bounds	Observed min	Observed max	Box used (%)	Normalised SD
Simulated energy proxy (kWh/100 km)	0 to 150	16.0	64.6	32.4	0.060
Task load	0 to 100	0	90.0	90.0	0.192
Informativeness	0 to 100	0	100.0	100.0	0.204
Pleasantness	0 to 100	5	100.0	95.0	0.212
Legibility	0 to 100	0	100.0	100.0	0.213

Table 10: Parameters of the recommended configurations for the 16 participants: between-participant dispersion, search variation within the five optimisation rounds, and retention in the Round 21 configuration. The between-/within-participant decomposition is descriptive and does not attribute all differences to stable preferences.

Parameter	M	SD	Range	Within-SD	Ratio	Retained
Leaf gauge	0.56	0.31	1.00	0.17	1.82	11/16
Numeric score	0.37	0.21	0.70	0.16	1.34	5/16
Framing element	0.45	0.29	0.97	0.18	1.64	7/16
Speedometer	0.48	0.33	0.92	0.19	1.73	cannot hide
Acceleration pictogram	0.52	0.27	0.94	0.17	1.53	8/16
Vehicle markers	0.51	0.32	0.96	0.17	1.89	8/16
Opacity	0.43	0.32	1.00	0.17	1.82	n/a

A label-permutation test further showed that the 93 Pareto designs were closer to others carrying their original participant label than under random relabelling (observed separation +0.090, 5,000-permutation $p = .0002$). The procedure tested association with

the label but did not reconstruct each Sobol seed and adaptive search path under permutation, so participant preference remains inseparable from path differences. In the random-slope models, the associations of the framing element with pleasantness and task

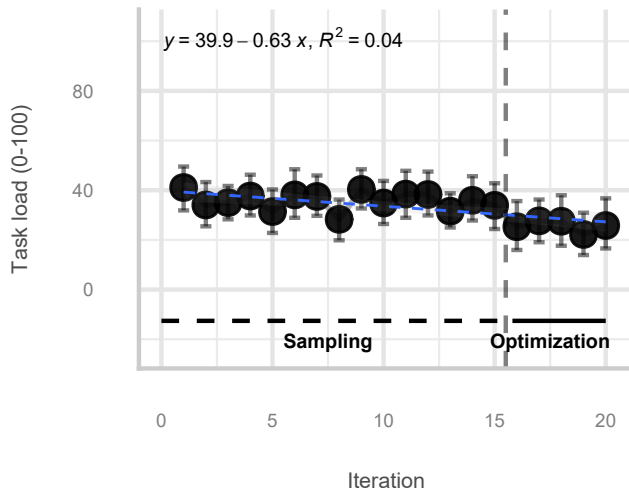


Figure 14: Task load across Rounds 1–20. Filled points with error bars show the participant mean and 95% t confidence interval for each round; the solid line is a descriptive linear trend across all 20 rounds, and the vertical dashed line marks the transition from sampling to optimisation after Round 15. The final repeat drive in Round 21 is excluded. The trend cannot distinguish practice, fatigue, rating drift, and the selection mechanism.

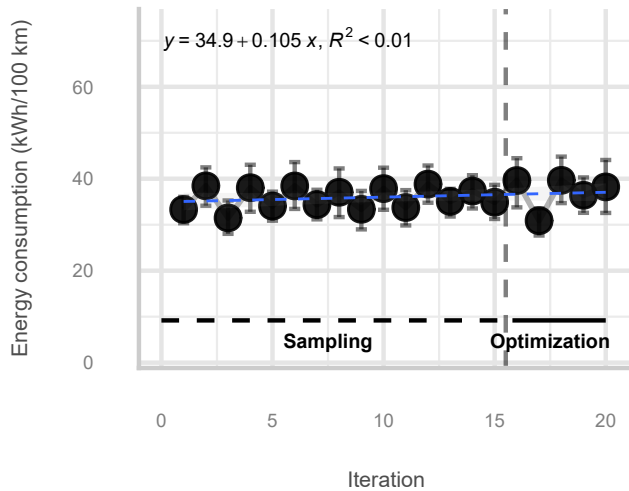


Figure 15: Simulated energy-intensity proxy across Rounds 1–20. Filled points with error bars show the participant mean and 95% t confidence interval for each round; the solid line is a descriptive linear trend across all 20 rounds, and the vertical dashed line marks the transition from sampling to optimisation after Round 15. The final repeat drive in Round 21 is excluded.

load showed between-participant heterogeneity, both with Holm-corrected $p < .001$; no other parameter-objective combination passed correction.

Open-ended responses also included opposing evaluations of the framing element: four participants described it as annoying or distracting, while another listed it as the favourite element. This is consistent with the heterogeneity models but remains participant-associated evidence from a small sample and requires retesting across days to confirm stability.

5.3.4 Is a Participant's Own Design Better for That Participant?

Determining whether a person's configuration is better than other people's configurations requires predicting responses to designs they have never experienced. The median leave-one-round-out R^2 of the per-participant ridge-regression response surfaces was -0.067 , and only 7/16 values were above zero; reducing the model to three parameters did not improve performance (-0.081). A negative value means that the model predicted the held-out round no better than the participant's own mean, so the subsequent transfer results are not confirmatory.

Given this limitation, the transfer analysis is presented only as an exploratory result. On average, the models assigned higher predicted scores to participants' own designs than to other participants' designs and to the common design selected by the group model. However, each participant's own design was also closer to their training data, and the underlying response surfaces performed poorly under cross-validation; adding a distance covariate did not remove the dependence on the model. These values can therefore generate hypotheses only and do not establish a transferable advantage of a personalised design for its owner.

A related property of the objective set affects how hypervolume should be interpreted. Within participants, the five objectives were correlated: task load and the three acceptance constructs loaded together on the first principal component, accounting for 50.7% of variance; energy loaded almost alone on the second, accounting for 20.1%; and only the first component survived a parallel analysis against column-wise shuffled data. The effective dimensionality was 3.02 out of five. Energy was the only objective that was not another facet of the acceptance construct, and its within-participant correlations with the other four objectives ranged from -0.01 to $.04$.

5.4 Feasibility, Acceptability, and Comparison of Motion Conditions (RQ3)

5.4.1 Session Duration and Completion. All 16 participants completed all 21 rounds. The median duration of the in-simulator portion was 24.2 minutes, with a range of 16.5–38.0 minutes. Longer rounds occurred mainly early in the session and were often accompanied by very low speeds and repeated contacts, which is consistent with participants becoming familiar with the steering wheel, pedals, and simulator vehicle. The 'collision count' in the logs comprises collision-entry events recorded by CollisionLogger: entries for the same collider were rate-limited within two seconds, normal rolling contact with the terrain was excluded, but another collision after the cooldown could be counted separately. It is neither a count of physical contact frames nor a basis for diagnosing a system fault. Sessions recorded 12–127 events (median 56). The maximum energy-proxy value in the formal experiment was 64.61 and did not reach the upper bound of 150.

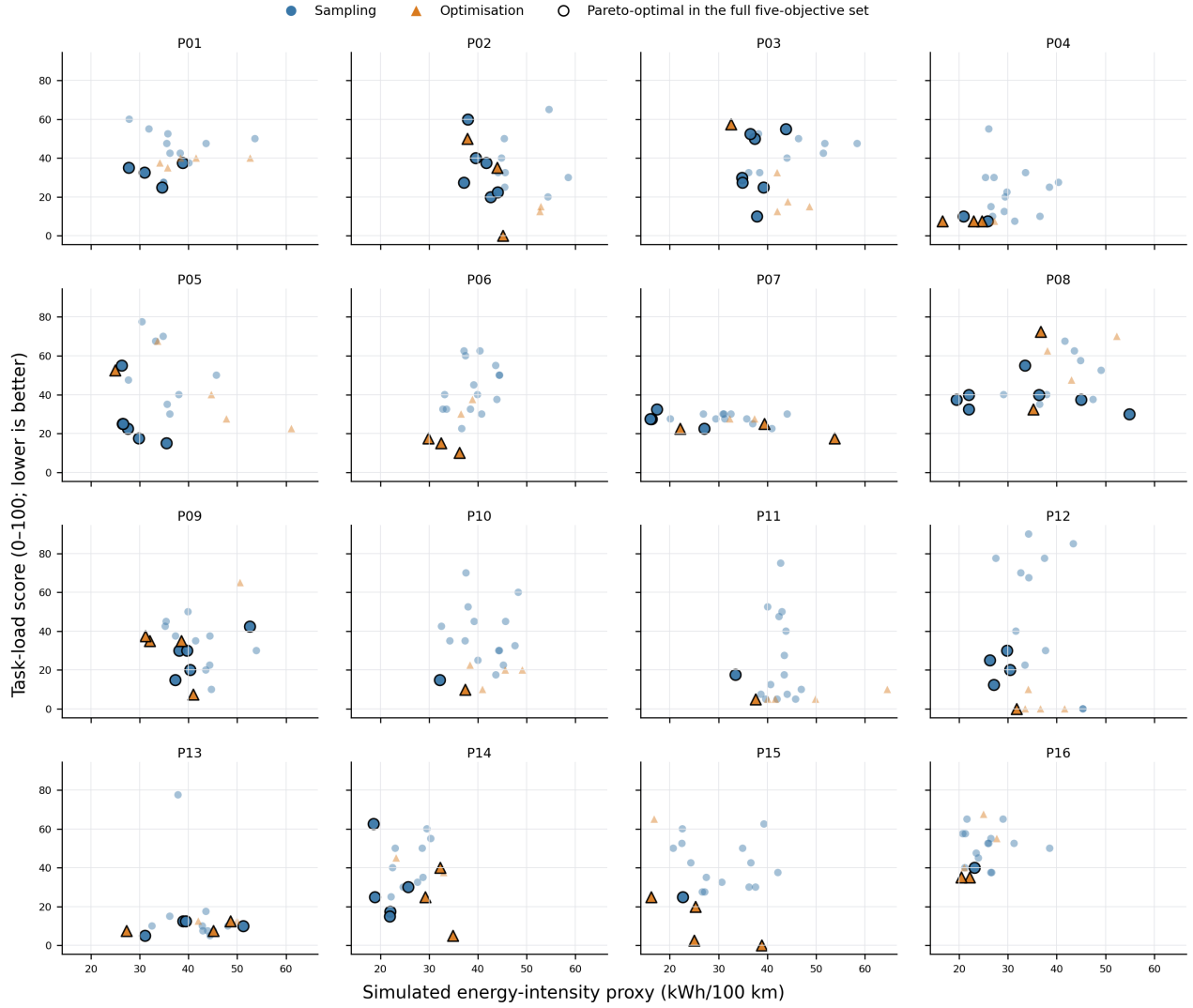


Figure 16: Two-dimensional projection of each participant’s designs from the first 20 rounds onto simulated energy-intensity proxy and task load. Sampling and optimisation rounds use different symbols; colour highlights points classified as Pareto-optimal using all five objectives. Axes are fixed across panels. Occlusion and relative position in the two-dimensional projection are not equivalent to five-dimensional dominance, and no connecting lines imply a trajectory.

5.4.2 Comparison of Motion and No-Motion Conditions. Because the motion condition changed over time after the eighth participant, the following analysis is exploratory only. Median hypervolume gains were 4.40% in the motion group and 14.29% in the stationary group ($U = 24$, $p = .442$, rank-biserial correlation = $-.25$, bootstrap 95% CI for the between-group median difference $[-27.46, 31.09]$). Median changes in Pareto hit rate were .0667 and .300, respectively ($U = 21$, $p = .279$, $r = -.344$, 95% CI $[-.6, .2]$). The p values for the five final same-route changes ranged from .328 to .959. After Holm correction across the seven sensitivity tests, all adjusted p values were 1.000, and all confidence intervals for differences crossed zero.

With only eight participants per group, non-random conditions, and complete overlap with recruitment order, the results neither show a clear device effect nor establish equivalence between the conditions.

5.4.3 Post-Study Questionnaire. All eight participants in the motion group rated the seat movement as ‘just right’.

On the five-point post-study scales, ‘the final design suited me’ had a mean of 4.38 ($SD = 0.72$), ‘the display changed towards my preferences across rounds’ 4.12 ($SD = 0.81$), ‘I would use such a display in a real car’ 3.94 ($SD = 1.12$), and perceived distraction

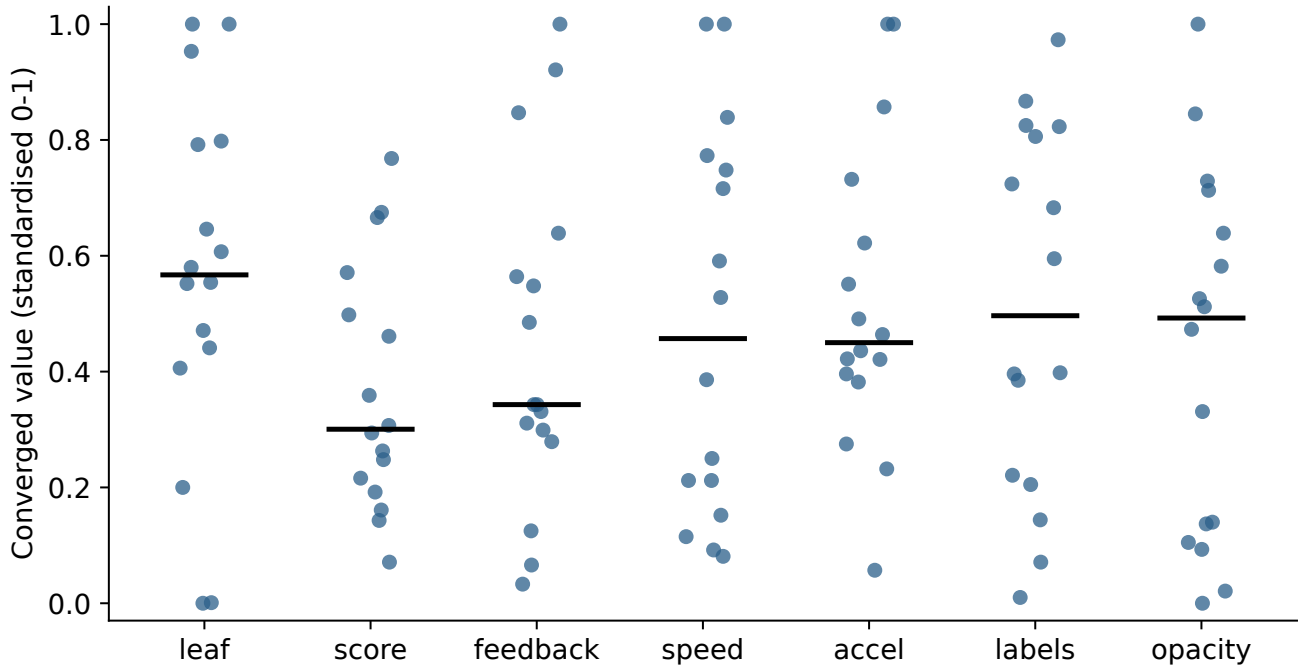


Figure 17: Values of the seven design parameters in the recommended Round 21 configurations. Each marker represents one participant; solid horizontal lines show medians.

2.00 ($SD = 1.21$). The item 'the final display helped me drive more efficiently' had a mean of 4.31 ($SD = 1.08$), but participants' judgements were influenced by the visible eco-score feedback in the interface and should therefore be interpreted as self-reported perceived helpfulness. Mean motion discomfort was 2.69 ($SD = 1.45$), and five participants gave a rating of 4 or 5.

Open-ended responses mirrored the parameter results, including their differentiation. Eight participants listed the leaf gauge as their favourite element, and one described an ideal display containing only a single leaf. The framing element received the most criticism: four participants called it annoying or distracting, and one said that the numerical score changed too quickly and caused dizziness when watched. The same elements nevertheless had advocates. One participant in the no-motion group preferred the advice text because it explained how to improve; another requested that the vehicle markers be retained; and another praised the colour-changing speedometer. Twelve of the 16 participants said that the final design was, or was close to, their ideal design.

5.5 Summary of Results

Overall, the fixed-order experiment showed that cumulative hypervolume increased between Rounds 15 and 20 for 15/16 participants, and that both the five-round-window comparison and Pareto hit rate moved in the direction favourable to the model-driven phase. The four subjective outcomes were associated with phase, whereas the simulated energy proxy showed no significant phase difference. The observed configuration selected for Round 21 improved

subjective ratings relative to the same-route sampling reference, although this comparison remained affected by order, practice, and rating drift. Configurations showed participant-associated differentiation, while the transfer analysis was exploratory because its cross-validated R^2 was negative. All participants completed the protocol; the device-condition sensitivity analysis was inconclusive.

6 Discussion

6.1 What Within-Session Optimisation Can and Cannot Establish

Each participant provided only 20 modelling evaluations for a seven-dimensional, five-objective search, of which 15 were allocated to space filling and five to model-driven selection. This budget can produce only a sparse Pareto approximation and cannot establish convergence. It also means that RQ1 should be understood as a description of a specific experimental workflow; the independent effect of the optimiser relative to random search, a fixed default display, or natural practice lies beyond what this design can estimate.

The five model-driven iterations occurred at the end of the session. Any changes in hypervolume, ratings, or behaviour over time were therefore simultaneously exposed to practice, fatigue, changes in rating strategy, and the route cycle; drift in the feedback users give over time is a recognised concern in interactive machine-learning systems [10]. The increase in cumulative hypervolume and the five-round-window comparison can be used to check whether the workflow produced a signal in the expected direction, but are insufficient on their own to identify a causal mechanism.

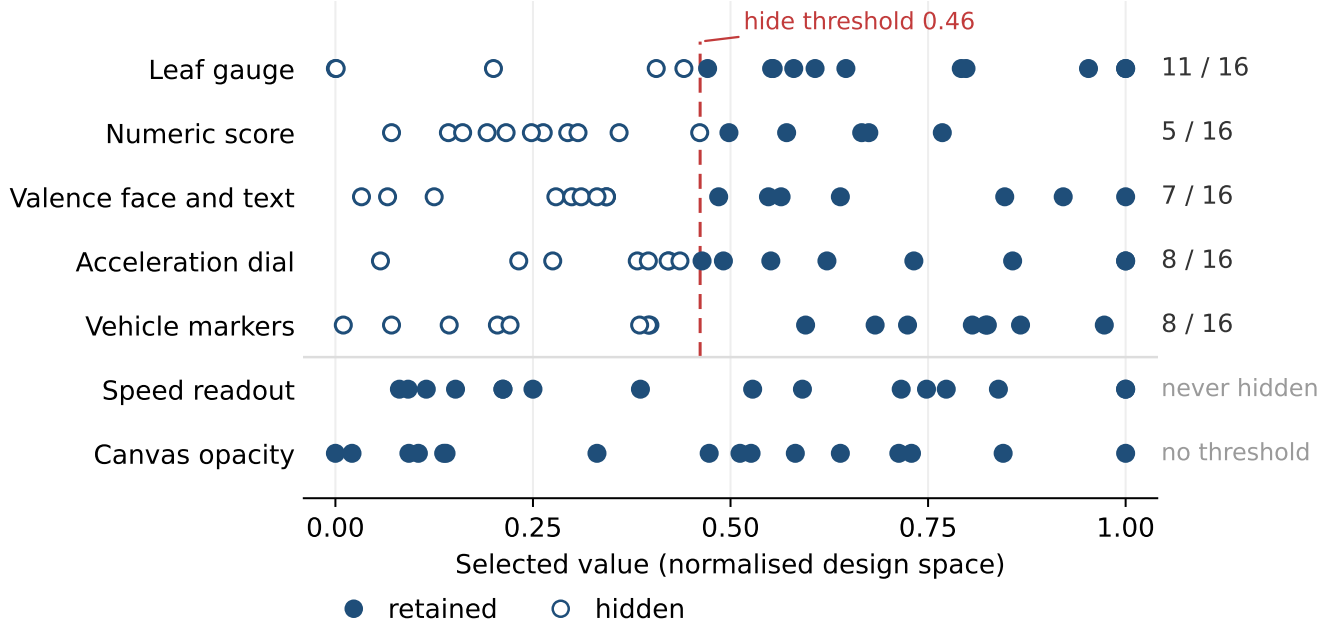


Figure 18: Normalised parameters of the recommended Round 21 configurations for the 16 participants. Filled points indicate that an element was shown and open points that it was hidden. The dashed line is the normalised hiding threshold of $0.6/1.3 \approx 0.462$, derived from the physical scale threshold in the implementation. The speedometer cannot be hidden and opacity does not use this threshold; both are shown below the separator.

Cumulative hypervolume increased for 15/16 participants, with a median gain of 10.9%; the gain over the five optimisation rounds exceeded that over the preceding five sampling rounds ($V = 107$, $p = .044$). Because cumulative hypervolume cannot decrease by construction [8] and the two windows occurred in a fixed sequence, these values support only the weaker statement that ‘the workflow continued to expand the observed front within a limited budget’.

The increase in Pareto hit rate from 0.24 during sampling to 0.44 during optimisation is also consistent with the expectation for model-driven selection. However, the front was defined retrospectively from all five objectives and all 20 rounds, four of the objectives came from participant ratings, and phase order remained fixed. The hit rate therefore provides a second form of descriptive evidence but does not remove order effects or rating drift.

The four subjective outcomes were more favourable later in the session, whereas the simulated energy proxy showed no significant phase change. This asymmetry is consistent with the difficulty, noted in earlier simulator work, of obtaining realistic energy-consumption signals from a driving simulator [3]. The most conservative interpretation is that, in this workflow, participant ratings entered the optimisation loop directly and could produce detectable phase associations relatively quickly, whereas the energy proxy followed a longer measurement chain through driving behaviour and extrapolation from short routes. The present design cannot determine how much of the subjective change arose from model-driven selection, nor can the null energy result be generalised to real vehicles.

Changing the energy bounds post hoc from $[0, 150]$ to $[15, 65]$ strengthened the internal comparison, showing that acquisition results were sensitive to the scale used to normalise the objective. Because the new bounds were selected from the same data, this was a sensitivity analysis rather than an independent validation of an alternative setting. Replications should set bounds using an independent pilot or a preregistered rule.

The energy proxy differed systematically by approximately 5.06 kWh/100 km between Round 21 and its source round. This cannot be reduced to measurement noise because the two drives occurred at different times in the session and may have involved different traffic and levels of familiarity. Longer routes, repeated retests, and randomisation of phase and route are required to separate measurement error from session-level change.

The association models attributed more variation in the energy proxy to driving-behaviour covariates and less to display parameters, but these shares depended on the model specification and constitute only an associative decomposition. The display may act indirectly through driving behaviour, and the model may omit route, traffic, and temporal factors. The study therefore reports model associations only and does not use them to establish either that the display is unrelated to energy or the opposite.

The conclusion for RQ1 is limited to this fixed-order workflow: subjective and Pareto signals consistent with model-driven selection emerged within a realistic evaluation budget. In the absence of an independent non-adaptive condition, however, the study cannot determine whether the method outperforms a carefully designed

fixed display. The most direct follow-up should randomly interleave optimisation with matched non-adaptive selections and include a strong fixed default.

6.2 Why Personalised Displays Differed

The observation under RQ2 that does not depend on an average phase improvement is the differentiation of the recommended configurations and Pareto sets under participant labels. However, each participant also received different initial Sobol points and followed a different adaptive path, so these differences combine preference, random seed, path dependence, and finite-sample error. They are participant-associated design differences; whether they are stable individual preferences remains to be confirmed through retesting across days.

Six of the seven dimensions covered nearly their full ranges, between-participant dispersion exceeded variation within the five optimisation rounds, and label permutation also showed that Pareto sets clustered by participant. The permutation retained the observed points but did not rerun an identical search path for each label, so it could not separate 'who provided the ratings' from 'which candidates were seen'. The random-slope results and opposing open-ended responses provide complementary evidence of heterogeneity, but the sample remains small.

'Different designs' and 'a person's own design is better for that person' are distinct claims. The transfer analysis had a median leave-one-round-out R^2 of -0.067, with only 7/16 values positive, showing that individual response surfaces predicted unseen designs poorly. The transfer rankings and personalisation gain can therefore only generate hypotheses. Confirming a benefit of personalisation requires optimisation repeated across days, participants experiencing other people's or a common configuration, and hierarchical or multi-task models.

Three explanations in previous work are consistent with this pattern. The first concerns use. Stillwater and Kurani [20] found that instantaneous performance feedback was used mainly to experiment with and learn new behaviours, while average-performance feedback was used mainly to set and achieve goals. The numerical score in my display was instantaneous. Under this account, disagreement about that element concerns whether a quantity for experimentation is still needed at that moment: drivers who have established a stable driving style no longer need it, whereas those still exploring do.

The second explanation concerns attention. Caution is needed here: Jamson et al. [12] did not find that more information was more distracting. They found that a more informative visual display reduced control error and that adding an auditory channel also reduced visual distraction. In other words, attentional cost depends on how information is distributed across sensory channels, not simply on how much information is present. My display concentrated all information in the visual channel, so opacity directly determined how much visual resource it occupied, and drivers differed in how much visual resource they were willing to allocate. The same opacity could feel appropriately present to one driver and as though it competed with the road for attention to another. This difference is an observation from the present study and has no corresponding conclusion in the existing literature. The third

explanation concerns representation: Dahlinger et al. [7] showed that numerical and symbolic presentations of the same information can produce different behaviour, suggesting that individual preferences between a numerical score and symbolic gauge are more than a matter of taste.

I offer a fourth explanation and explicitly label it as speculation. Elements of this display are partly redundant: the leaf gauge, numerical score, and framing element all report different aspects of the same underlying eco-score. An optimiser that minimises task load while maximising informativeness has an incentive to retain only one of them, and which one it retains may be determined by small differences in participants' early ratings. If so, part of the differentiation is a property of the design space rather than entirely a property of the driver. Distinguishing the two would require repeating the optimisation with the same participant on another day. I did not do this, so the explanation remains speculative.

Whichever explanation applies, the practical implication is consistent with the conclusions reached from different directions by Brouwer et al. [3] and Sumner et al. [21]: fixing one display for all drivers removes a genuine source of variance.

Open-ended responses made the differentiation concrete. Eight participants selected the leaf gauge as their favourite, and one wanted only a single leaf. Four described the framing element as annoying or distracting. Yet another participant preferred it precisely because the advice text explained how to improve; one requested that the vehicle markers be retained; and another praised the colour-changing speedometer. One driver removed an element that another would use the display specifically to obtain. This is a direct observation underlying the individual-differences claim.

6.3 Designing for a Constrained Evaluation Budget

Two design decisions were made specifically to protect the evaluation budget and are worth reporting because they are transferable. First, element presence was folded into element size, keeping the entire search space continuous. A 14-dimensional mixed discrete-continuous space would be much harder to search in 20 evaluations than a seven-dimensional continuous space, and a mechanism for modelling the mixed space would introduce assumptions that a small sample could not test. The cost is that the optimiser cannot distinguish an element that was intentionally removed from one that was merely made very small. Dudley and Kristensson [10] note that representational choices of this kind in interactive machine-learning systems shape what the system can learn.

Second, acceptance was separated into three objectives. Averaging them would reduce the number of objectives from five to three and make optimisation easier, and this was the approach taken by Jansen et al. [14], who combined two acceptance items into one objective. I kept the items separate because informativeness, pleasantness, and legibility could move in opposite directions after the same design change; averaging would conceal the very conflict that motivates the optimiser. Whether this trade-off was correct remains an open question, because the study did not run an averaged-objective variant.

The protocol itself - 21 rounds in a head-mounted display - approaches the upper end of a reasonable range. Previous human-in-the-loop interface studies used similar per-person budgets of 15–40 evaluations [4, 14], so the burden was not unusual for the method, but it was unusual for a driving-simulator study; this combination warrants attention in any replication. There is also an experiential cost shared by the protocol and the method. Chan et al. [4] found that optimiser-guided participants reported lower agency and expressiveness and felt less able to control the process even though the optimiser improved their coverage of the design space and solution quality. Participants in the present study had no control over which design appeared next, so the loss of agency identified in that setting was present here in a stronger form.

6.4 Implications for Eco-Driving Display Design

If the observed differentiation is real, the design question shifts from 'which single display should leave the factory?' to 'which parts of the display should be adjustable, and who should adjust them?'. The results address this question concretely because the seven dimensions differed in the extent of their differentiation. Parameters on which participants converged are candidates for fixed design decisions; parameters on which participants disagreed are candidates for user control or adaptation.

Vehicle markers, the leaf gauge, and overall opacity showed the greatest differentiation in the data (ratios of 1.82–1.89). The numerical score showed the least (1.34) and was also the least frequently retained element, appearing in only five of the 16 final designs. Whether eco-information is expressed numerically or symbolically has itself been shown to change fuel consumption in the field [7], so low retention of the numeric channel is a substantive design signal.

This is a more restrained conclusion than 'complete per-driver optimisation' and is also more deployable. A production system cannot require 20 instrumented drives with a Bayesian optimiser before configuring the dashboard. It could, however, expose two or three settings, and the value of a study of this kind is in identifying which two or three they should be. This supports, from another direction, the position of Tu et al. [23]: adoption of eco-driving guidance depends on whether it fits the driver.

A further implication concerns what the display is for. I operationalised the usefulness dimension of acceptance as informativeness because fuel consumption could not be perceived directly in the simulator and behaviour change was not something participants could rate immediately. This is a defensible adaptation, but it means that the acceptance objective measured how well the display informed; persuasiveness was not measured. A display optimised to inform is not necessarily one optimised to change behaviour. Sanguinetti et al. [17] likewise separate design dimensions from behavioural mechanisms: information, timing, and presentation map to attention, motivation, and learning, respectively, so 'what the display communicates' and 'what behaviour the display produces' are not the same question in that framework.

6.5 Ethics and UK AI Governance Principles

This section introduces the five cross-sector AI regulatory principles proposed by the UK Government [9]. The framework emphasises understanding risk in context; the discussion below uses it to describe the system's research controls, residual risks, and deployment requirements.

6.5.1 Safety, Security, and Robustness. The experiment took place in a virtual environment and therefore created no risk of a real road collision, but it did create risks of virtual-reality discomfort, visual-vestibular conflict, and mechanical hazards from the motion platform. The participant information sheet and the research team's risk assessment specified measures including stopping, resting, and limiting device travel. Route guidance was entirely excluded from the design space, and the speed display retained minimum size and readability levels; these were constraints of the study design. The platform failure during the study also shows that robustness cannot be represented by mean frame rate alone. Deployment would additionally require fault detection, safe fallback, latency measurement, and validation on real roads.

6.5.2 Appropriate Transparency and Explainability. The participant information sheet stated that the display would be adjusted according to participants' ratings, but did not explain the surrogate model, acquisition function, or generation of the next configuration in detail. This general disclosure was sufficient to tell participants that the system would adapt, but may have produced expectation effects and would not support informed use in deployment. A driver-facing version should explain what data are collected, which objectives are optimised, why the current configuration has changed, and the system's uncertainty and limitations, in a form that does not increase distraction while driving.

6.5.3 Fairness. Personalisation can be fair only if the design space and training data cover different needs. The current sample was small (16) and concentrated in age, and excluded several health conditions; it cannot assess differences by age, disability, driving experience, or culture. Size bounds and hiding thresholds also exclude needs outside those boundaries. Future work should define accessibility constraints in advance, report subgroup performance, and avoid imposing longer cold starts or less stable configurations on sparsely represented groups.

6.5.4 Accountability and Governance. A per-participant system requires configuration-level logs to reconstruct what someone saw in a given round, which data entered the system, which model version was used, and why the configuration was selected. This study stored the seven-dimensional parameters and five objectives, but the scope of public release remains constrained by the consent materials and re-identification risk. A deployed system would also need to identify the data controller, the person approving model updates, processes for reviewing incidents and anomalies, a version-rollback mechanism, and ultimate human responsibility. The optimisation tool itself cannot be the accountable party.

6.5.5 Contestability and Redress. Participants could stop the complete study at any time, but the experimental workflow provided no round-level control to skip the current configuration, immediately return to the previous one, or reset the personalisation model;

ratings could affect only subsequent rounds. This limitation had a methodological rationale in the study protocol but should not be transferred directly to a product. A deployed version should provide clear means to leave personalisation, pause learning, restore a safe default, withdraw data, and contact human support, while ensuring that these choices do not reduce the availability of basic driving information.

6.5.6 Adaptivity and Autonomy of the System. The system refitted its model after each round using new ratings and the energy proxy, demonstrating adaptivity; the acquisition function proposed the next configuration without a researcher specifying its parameters round by round, demonstrating limited autonomy. This autonomy was bounded by the predefined seven-dimensional limits and five objectives, and ultimate research responsibility remained with the researchers. Assessing risk in context is more meaningful than assigning the system a single categorical label.

6.5.7 Privacy, Social Harm, and Misuse. Per-participant optimisation continuously collected driving trajectories and subjective ratings. Even after names are removed, such data may permit re-identification through behavioural patterns. The existing consent materials contain an ambiguity that must be resolved regarding the scope of public data, so high-dimensional trajectories must not be released in advance. Foreseeable misuse includes applying the same loop to prolong glances, increase engagement with advertising, or optimise other objectives that conflict with drivers' interests. Mitigations include data minimisation, purpose limitation, independent safety constraints, prohibiting commercial engagement from replacing safety objectives, access controls, and a re-identification assessment before release.

6.6 Limitations

The most important limitation is the absence of a non-adaptive baseline. Within each participant, the study compared optimisation designs with quasi-random samples from the same space. The term baseline refers to this internal, route-matched comparison: the participant's own sampling round on the same route. It is not the fixed, carefully designed default display that a manufacturer might deploy, and the study included no such condition. All claims in the thesis are limited accordingly.

The measurement instruments were also simplified. Two items replaced the six subscales of NASA-TLX [11], and three brief items replaced the complete acceptance scale [24]. The reliability of using one item to measure a multi-item construct is unknown, and a single-item scale cannot provide internal-consistency estimates. The results apply only to the questions that were asked and cannot be generalised to the validated constructs from which those questions were adapted. Jansen et al. [14] made the same shortcut for the same reason, making it conventional in this class of study but not cost-free.

A more fundamental limitation concerns the simulator's ecological validity. Participants drove in a head-mounted display with a motion platform, and harsh acceleration or speeding had no real consequences. Fuel consumption is especially important: it was imperceptible to drivers in the simulator but can be perceived in a real vehicle. One prior result makes this more than a generic

caution. Pietra et al. [16] compared haptic and visual eco-driving feedback with ten participants in a virtual-reality simulator and found that haptic feedback reduced the mean power index by 14%, the combined condition reduced it by 11%, and visual feedback alone increased it by 11%. The present display was purely visual and was evaluated in virtual reality, so that study is the closest published comparison and points in an unfavourable direction. I cannot exclude the possibility that the energy objective in this study was affected by the same phenomenon.

Performance logs recorded approximately 73,245 frames over 16.96 minutes during partially overlapping periods, with a mean of 71.99 fps. This shows that the sampled period was close to the 72 Hz ceiling, but it is not an end-to-end motion-to-photon latency measure and did not cover every session. The high discomfort ratings reported by five participants therefore cannot be attributed exclusively to, or used to exclude, any single explanation among dropped frames, latency, virtual-reality presentation, motion cues, or visual-vestibular conflict.

Route and phase were not fully separated. The four routes cycled in a fixed order; the route with the highest driving difficulty appeared twice in the five optimisation rounds and three times in the 15 sampling rounds, and the final recommended configuration was also always retested on the same route. The available evidence does not justify describing that route as the 'slowest' or as having 'the most junctions'. Randomising or balancing route order could reduce this confound.

The same design introduced order effects. Each participant underwent optimisation once, within one sitting, and the design sequence depended on their previous ratings. Familiarity with the road network, vehicle control, and the rating task all increased monotonically during the session and were therefore confounded with the transition from sampling to optimisation. This is an inherent problem for a single-session human-in-the-loop protocol.

Rating behaviour changed with round number: participants completed questionnaires faster, used extreme ratings more often, and changed some relationships between items. Because phase was defined by round, these changes could not be separated from model-driven selection. Pareto hit rate does not average ratings into one level, but its five-dimensional dominance judgement still uses four rating objectives directly, so it cannot be described as independent of rating behaviour. Randomly interleaving sampling and optimisation rounds and examining rating stability in retesting across days would provide a stronger design.

The objective set itself was also mismatched, with a lower effective dimensionality than assumed by hypervolume. Within participants, task load and the three acceptance constructs loaded together on one component that accounted for half of the variance, while energy loaded almost alone on the second; only the first component survived parallel analysis. The effective dimensionality was 3.02 out of five. Hypervolume was nevertheless calculated across all five axes, allowing the optimiser to receive credit for dimensions carrying little independent information and making absolute hypervolume less meaningful than its within-participant change. The implication for the objective set is two-sided: it supports combining the four subjective objectives into fewer axes in a replication, but also supports retaining energy because it is the only objective that is not another facet of acceptance.

The sample was concentrated in age and had limited driving and simulator experience, restricting generalisation. The equipment change after the eighth participant also completely overlapped with temporal order. All seven Holm-adjusted *p* values in the exploratory sensitivity analysis were 1.000, with wide confidence intervals crossing zero; the available data therefore leave the equipment effect uncertain.

7 Conclusion

This study explored eco-driving display design as a per-participant, human-in-the-loop multi-objective search problem. Across seven continuous display parameters, the system used five objectives: an energy-intensity proxy calculated from simulated road load, task load, and three acceptance ratings. Each participant completed 15 Sobol sampling rounds, five model-driven optimisation rounds, and one retest of an observed recommended configuration.

In a single session with sampling fixed before optimisation, cumulative hypervolume increased for 15/16 participants and the Pareto hit rate was higher during optimisation than during sampling. The four subjective outcomes were associated with phase, whereas the simulated energy proxy showed no significant phase difference. Recommended configurations improved subjective ratings relative to the same-route sampling reference and differed markedly between participants. However, phase order, practice, rating drift, different Sobol seeds, and search paths jointly limit interpretation. The present results cannot establish that the optimiser caused the improvement, that stable individual optima converged, or that energy use would decrease in real vehicles.

Beyond these associations, the study contributes an inspectable measurement chain: a driveable reproduction of a real urban district, a display whose element sizes and overall opacity form a continuous design space, and an optimisation loop that fits within a single session. Logging links every rating to the exact configuration and driving trajectory that produced it. The recommended configurations also carry a design-relevant observation. Participants differed most in which elements were retained at all, and several recommended designs kept only a subset of the available channels. Subject to the caveat that search paths contributed to this differentiation, this points towards element presence, rather than fine size tuning, as the first-order personalisation decision for eco-driving displays.

The principal value of this study is therefore a system implementation that can be inspected and a set of empirical observations about limited budgets, measurement chains, and participant-associated differences. The most direct follow-up should randomly interleave model-driven and non-adaptive selections, include a strong fixed default, retest the same participants on different days, and validate behavioural effects using longer routes or real-vehicle energy measurements. The released source code, together with the data and analysis materials whose release is pending ethics review, is intended to make that comparison straightforward to build.

Open Science and Materials Availability

The public repository at <https://github.com/ArcadiaE/Sustainable-Personalised-Driving> provides the self-authored Unity/Python

source code, the releasable pseudonymous study data, the R analysis scripts with their outputs, this dissertation, and documentation. Its README states that Cesium, CityGen3D, and other third-party assets are not distributed with the repository. This manuscript does not claim that 'all data and the complete project are openly available'.

The released set comprises the per-round master table, questionnaire-derived measures, per-participant optimiser observation and hypervolume logs, study configuration with per-file script hash manifests, the analysis scripts, and a data dictionary; this is the minimum required to reproduce the main statistical results. Raw post-session questionnaire exports (demographic combinations and free text), continuous high-frequency driving trajectories, detailed console logs, and the verbatim per-session script snapshots are withheld pending a re-identification-risk assessment. If the finally approved scope permits access only by authenticated researchers, the data will be supplied under that governance condition rather than misdescribing pseudonymisation as anonymisation.

Before submission of the final English manuscript, a version tag and commit hash will be frozen for the code and approved analysis materials. The record will include versions of Unity, Python/R, BoTorch, and key dependencies, random seeds, execution commands, licences, and the list of excluded third-party assets. An immutable archive and DOI will be created through Zenodo or the Open Science Framework. No DOI has yet been issued, so no invented identifier is entered here.

Generative AI Use Statement

This study used Anthropic Claude and OpenAI Codex as assistive tools. Their uses included learning and configuring virtual-reality equipment, the motion platform, and other hardware, as well as Unity and other software environments. They also assisted with developing, debugging, and checking simulator and analysis code; literature searching and verification of reference metadata; scripts for generating statistical figures; and proofreading, terminology harmonisation, and language polishing of the thesis. During preparation of the final English LaTeX manuscript, they were also used for format conversion and mechanical consistency checks against the submission template.

All participant data were collected by the author under the approved protocol; AI did not generate, replace, or modify any observation. The research design, methodological choices, interpretation of results, research content, argumentative structure, and principal text of the thesis were written by the author. The author reviewed each AI suggestion and decided whether to adopt it, and verified all citations, numerical values, and statements used in the final text against the original materials. The author accepts responsibility for the complete research project and thesis.

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A Software Environment

All analyses were run in R 4.6.1 on Windows 11. The complete sessionInfo() output is reproduced below so that the package versions behind every reported number are on the record.

```
R version 4.6.1 (2026-06-24 ucrt)
Platform: x86_64-w64-mingw32/x64
Running under: Windows 11 x64 (build 22631)

Matrix products: default
LAPACK version 3.12.1

locale:
[1] LC_COLLATE=Chinese (Simplified)_China.utf8
[2] LC_CTYPE=Chinese (Simplified)_China.utf8
[3] LC_MONETARY=Chinese (Simplified)_China.utf8
[4] LC_NUMERIC=C
[5] LC_TIME=Chinese (Simplified)_China.utf8

time zone: Europe/London
tzcode source: internal

attached base packages:
[1] stats      graphics  grDevices  utils      datasets  methods   base

other attached packages:
[1] emoa_0.5-3      moocore_0.3.2      rmcrr_0.7.0      patchwork_1.3.2
[5] ggstatsplot_1.1.0 ARTool_0.11.2      effectsize_1.0.3  performance_0.17.1
[9] parameters_0.29.2 emmeans_2.0.4      lmerTest_3.2-1    lme4_2.0-6
[13] Matrix_1.7-5    purrr_1.2.2        tidyr_1.3.2      dplyr_1.2.1
[17] colleyRstats_0.1.6 ggplot2_4.0.3

loaded via a namespace (and not attached):
[1] Rdpack_2.6.6      mnormt_2.1.2      polynom_1.4-1
[4] gridExtra_2.3.1   rematch2_2.1.2    sandwich_3.1-3
[7] rlang_1.3.0        magrittr_2.0.5     multcomp_1.4-32
[10] PMCMRplus_1.9.12  otel_0.2.0         matrixStats_1.5.0
[13] compiler_4.6.1    statsExpressions_2.1.0 BWstest_0.2.3
[16] vctr_0.7.3        kSamples_1.2-12    quantreg_6.1
[19] stringr_1.6.0     pkgconfig_2.0.3    fastmap_1.2.0
[22] backports_1.5.1   labeling_0.4.3     rmarkdown_2.31
[25] nloptr_2.2.1      MatrixModels_0.5-4 xfun_0.60
[28] cachem_1.1.0      Rmpfr_1.1-2        SuppDists_1.1-9.9
[31] gmp_0.7-5.1       psych_2.6.5        parallel_4.6.1
[34] cluster_2.1.8.2   R6_2.6.1           stringi_1.8.9
[37] RColorBrewer_1.1-3 rpart_4.1.27       car_3.1-5
[40] boot_1.3-32       numDeriv_2016.8-1.1 estimability_2.0.0
[43] knitr_1.51        Rcpp_1.1.2         zoo_1.9-0
[46] DHARMA_0.5.0     base64enc_0.1-6    correlation_0.8.8
[49] nnet_7.3-20       splines_4.6.1      tidyselct_1.2.1
[52] rstudioapi_0.19.0 dichromat_2.0-1    abind_1.4-8
[55] codetools_0.2-20 lattice_0.22-9     tibble_3.3.1
[58] plyr_1.8.9        withr_3.0.3        bayestestR_0.18.1
[61] S7_0.2.2          evaluate_1.0.5     coda_0.19-4.1
[64] foreign_0.8-91    survival_3.8-6     ggpp_0.6.1
[67] RcppParallel_6.2.0 confintr_1.0.2      pillar_1.11.1
[70] carData_3.0-6     stats4_4.6.1       checkmate_2.3.4
[73] reformulas_0.4.4  insight_1.5.3      generics_0.1.4
[76] paletteer_1.7.0   rstantools_2.7.0   scales_1.4.0
[79] minqa_1.2.8       xtable_1.8-8       glue_1.8.1
[82] Hmisc_5.2-6       tools_4.6.1        see_0.14.1
[85] ggpmisc_1.0.0     data.table_1.18.4  SparseM_1.84-2
[88] gsgsignif_0.6.4   mvtnorm_1.4-2      grid_4.6.1
[91] rbibutils_2.4.1   datawizard_1.3.1   colorspace_2.1-3
[94] nlme_3.1-169      htmlTable_2.5.0    Formula_1.2-6
[97] cli_3.6.6         viridisLite_0.4.3  gtable_0.3.6
[100] digest_0.6.39     prismatic_1.1.2    ggrepel_0.9.8
[103] TH.data_1.1-5     htmlwidgets_1.6.4  farver_2.1.2
[106] htmltools_0.5.9   memoise_2.0.1      lifecycle_1.0.5
[109] multcompView_0.1-12 MASS_7.3-65
```

B Post-Session Questionnaire

Table 11 lists the full content of this appendix.

The post-session questionnaire was administered once, after the final round, as a web form rather than inside the headset. It collected

the demographics reported in Section 3.2 as well as the items below. Items `design_1` to `design_5` shared the stem *How much do you agree with each statement about the FINAL display design?* and used a five-point scale from strongly disagree to strongly agree. All sixteen participants completed it. Variable names are those used in the released data.

Table 11: Items of the post-session questionnaire, with the variable names used in the released data.

Variable	Item
<code>pcode</code>	Participant code (entered by the experimenter, e.g. P01)
<code>age</code>	Your age (years)
<code>gender</code>	Your gender
<code>yearsdriving</code>	Years of driving experience
<code>drivefreq</code>	How often do you currently drive?
<code>vrex</code>	Experience with VR headsets
<code>simexp</code>	Experience with steering-wheel simulators
<code>design_1</code>	The final display helped me drive in a more eco-friendly way.
<code>design_2</code>	The final display distracted me from the road.
<code>design_3</code>	Over the rounds, the display changed towards a design that suited me better.
<code>design_4</code>	The final display design suited me.
<code>design_5</code>	I would want a display like this in a real car.
<code>discomfort</code>	During the session I felt dizzy, nauseous, or had eye strain (1 = not at all, 5 = very much)
<code>seatmotion</code>	The motion of the seat felt:
<code>open1</code>	Which display element did you like most, and which annoyed you most?
<code>open2</code>	Is the final design your ideal design? If not, what would your ideal display look like?

C Open-Ended Responses

Table 12 lists the full content of this appendix. Responses originally given in Chinese are presented in the author’s English translation, marked (tr.); the verbatim originals are preserved with the study materials.

The two free-text questions were *Which display element did you like most, and which annoyed you most?* and *Is the final design your ideal design? If not, what would your ideal display look like?* Responses given in English are reproduced verbatim, including spelling and grammar as written, and a dash marks a question left blank.

D Converged Design of Each Participant

Table 13 lists the full content of this appendix.

Each value is the normalised parameter in $[0, 1]$ that the optimiser proposed as that participant’s final design and that was then driven in round 21. The mapping from these values to element sizes and opacity is given in Section 3.4, and an element disappears once its size falls below the legibility threshold.

E Per-Participant Optimisation Progress

Table 14 lists the full content of this appendix.

The hypervolume (HV) columns are taken from the optimiser’s own log, computed in the normalised minimisation space with reference point $(1, 1, 1, 1, 1)$; *HV start* is the value after the fifteen sampling rounds and *HV end* the value after the last optimisation round. The Pareto columns are recomputed independently of the optimiser, a round counting as Pareto-optimal when no other round

Table 12: Verbatim open-ended responses from all sixteen participants.

P	Liked most / annoyed most	Ideal design
P01	leaf instruction	no I want leaf + instruction type
P02	the leaf	yes it is
P03	Like most: Leaf. Annoyed most: The face with the words under it.	Yes.
P04	Smile face	The opacity was slightly too high, though only for the smiley face (tr.)
P05	The leaf with red/green color, and the smile (crying) face with colors (of course the color change is my best preference)	Yes, the number does not change so quickly. So quick changes makes me dizzy and being distracted by the number (i prefer to watch the speed and i think the speed is the most important)
P06	The speed with its colour cue, and the changing face of the eco symbol (tr.)	yes
P07	I think the three-level colour design is good; the eco level can be seen at a glance (tr.)	The ideal design would keep only a single leaf, without other indicators such as speed (tr.)
P08	—	—
P09	The word comments below the Eco-Score display, it can help me know how to improve eco driving.	Nearly the ideal, but I want to keep other car’s label
P10	The low transparency, I’d like more information displayed on the front screen, but when they’re too much, it distracted me, so low transparency helped reduce that distraction.	Yes
P11	—	YES
P12	I found it quite intuitive that the colour changed on the main speedometer - made it easier to glance at only one thing while driving	Yes it is - I would have appreciated a bit more explanation on what the different components did on my screen because I felt like they changed at different rates or showed different things.
P13	—	—
P14	i like the leaf most, and i do not like the face as it might display negative emotion	i think it is, it’s just that i do not care a lot about eco driving but i probably consider using it if environment really needs me to do that
P15	leaf	no, the face emoji will distract me
P16	I liked the leaf in the ECO display, while the text annoyed me the most, as it distracted me from the road.	Yes, it is my ideal design

by the same participant is at least as good on all five objectives and strictly better on one. Rates are per round, over fifteen sampling and five optimisation rounds.

F Script Versions

Table 15 lists the full content of this appendix.

Every session archive carries a snapshot of the scripts that produced it, with modification time, size and hash. The table below is the snapshot taken for the first participant; the released data carry the corresponding per-file hash manifest for every participant, and the verbatim snapshots are preserved in the private archive pending re-identification review. The scripts that changed during data collection, and the sessions on either side of each change, are reported in Section 3.10.

Table 13: The seven design parameters of each participant’s final display.

P	Leaf	Score	Framing	Speed	Accel	Markers	Opacity
P01	0.953	0.666	0.066	0.773	0.382	0.595	0.582
P02	0.646	0.461	0.279	0.716	0.422	0.823	0.845
P03	0.792	0.294	0.343	0.25	0.232	0.806	0.526
P04	0.441	0.263	0.639	1	1	0.398	0
P05	0.554	0.143	0.921	0.528	0.551	0.867	0.713
P06	0.2	0.498	0.564	0.839	0.857	0.396	0.137
P07	0.406	0.248	0.847	0.212	0.622	0.144	0.512
P08	0.607	0.192	0.033	0.115	0.732	0.724	0.729
P09	1	0.216	1	0.152	0.491	0.01	0.639
P10	0.58	0.571	0.548	0.748	0.464	0.205	0.093
P11	1	0.307	0.299	0.081	1	0.071	1
P12	0	0.359	0.311	1	0.275	0.385	0.14
P13	0.798	0.675	0.343	0.092	0.057	0.221	0.105
P14	0.471	0.071	0.125	0.212	0.396	0.683	0.473
P15	0.001	0.161	0.485	0.591	0.421	0.825	0.021
P16	0.552	0.768	0.331	0.386	0.436	0.973	0.331

Table 14: Hypervolume and Pareto counts for each participant.

P	HV start	HV end	Gain	Gain			
P01	10.29	10.29	0.00	0.0	4	0.27	0.00
P02	13.88	23.05	9.17	66.1	9	0.40	0.60
P03	19.42	19.44	0.02	0.1	8	0.47	0.20
P04	21.22	22.41	1.19	5.6	5	0.13	0.60
P05	15.94	15.96	0.01	0.1	7	0.40	0.20
P06	8.24	12.34	4.10	49.7	3	0.00	0.60
P07	13.00	16.20	3.20	24.6	7	0.27	0.60
P08	5.04	5.20	0.16	3.2	9	0.47	0.40
P09	10.48	15.99	5.51	52.5	9	0.33	0.80
P10	15.57	17.80	2.23	14.3	2	0.07	0.20
P11	20.11	20.16	0.04	0.2	2	0.07	0.20
P12	24.95	25.92	0.97	3.9	5	0.27	0.20
P13	19.89	21.38	1.50	7.5	7	0.27	0.60
P14	7.95	10.74	2.79	35.0	8	0.33	0.60
P15	11.83	14.21	2.37	20.1	5	0.07	0.80
P16	7.01	8.01	1.00	14.3	3	0.07	0.40

Table 15: Simulator scripts at the time of the first session, with size and hash.

Script	Modified	Bytes	Hash
StudySetup.cs	2026-08-19 15:06	20028	09341EB50165
EcoFeedbackHUD.cs	2026-08-19 18:43	10554	92EE4A897C3B
EcoScore.cs	2026-08-19 12:47	9817	DA23A2FD08E6
RouteSet.cs	2026-08-19 12:47	3398	7BEDDD468CD9
RoundController.cs	2026-08-19 12:47	10089	C70A623FC4F9
StudyLogger.cs	2026-08-19 12:47	9824	B32A96F35A04
YawMotion.cs	2026-08-19 18:24	8725	31EAEA817920
YawDirectLink.cs	2026-08-19 12:47	13238	A8072504134E
CarController.cs	2026-08-19 12:47	30498	8BB6F6BA145F
BoForUnityManager.cs	2026-08-19 12:46	54139	C099F1D7C588

G Study Materials

This appendix contains the participant-facing study materials approved under UCL Research Ethics Committee project 1165: the task instructions as communicated to participants, followed by the original participant information sheet and consent form, reproduced unaltered on the following pages.

described in the participant information sheet. The task-relevant content communicated to every participant was as follows.

- You will drive short city rounds in virtual reality (approximately 22 in the information sheet; the realised protocol comprised 21), rating the display after each round.
- The study takes place at UCL and lasts approximately 40 to 60 minutes.
- The simulator records your driving (speed, acceleration, steering, and vehicle position), the resulting energy use, and your questionnaire answers.
- If you feel unwell, you will be asked to stop, remove the headset, and remain seated for a moment or get some air; the study is aborted on discomfort.
- Breaks are provided every 30 minutes or sooner, following the manufacturer’s usage guidance.
- Please do not mention your name or talk about private matters during the session.

G.2 Participant Information Sheet

The original participant information sheet (four pages) is reproduced on the following pages.

G.1 Task Instructions

There was no separate written task-instruction sheet. After giving consent and completing the demographic questionnaire, each participant received a short oral demonstration of the setup, as

Participant Information Sheet

UCL Research Ethics Committee Approval ID Number: __1165__

YOU WILL BE GIVEN A COPY OF THIS INFORMATION SHEET

Title of Study: Sustainable Personalised Driving

Department: UCL Interaction Centre, University College London

Name and Contact Details of the Researcher(s):

Mark Colley Email: m.colley@ucl.ac.uk

Yike Zhang Email: ucab338@ucl.ac.uk

Name and Contact Details of the Principal Researcher:

Mark Colley Email: m.colley@ucl.ac.uk

1. Invitation Paragraph

Before you decide whether to take part it is important for you to understand why we are doing this research and what participation involves. Please take time to read the following information carefully and discuss it with others if you wish. Ask us if there is anything that is not clear or if you would like more information. Thank you for reading this.

2. What is the project's purpose?

This study is being conducted by the University College London (UCL).

Mobility is a key aspect for inclusion. It is also of major importance for industry, and, unfortunately, a significant contributor to climate change. However, certain aspects have to be personalized. How a person drives strongly affects how much energy the same vehicle uses on the same road, and eco-driving displays are one of the cheapest ways to support this. However, a single fixed display design does not suit every driver.

We want to evaluate whether the design of an eco-driving head-up display can be personalised to the individual driver, online, within a single session. For this, we leverage Virtual Reality. You will drive a short city route repeatedly, and after each drive the display design is adjusted based on the ratings you give.

3. Why have I been chosen?

We are looking for people without dementia and severe illnesses. Participants who are pregnant, elderly, or have pre-existing binocular vision abnormalities, psychiatric disorders, heart conditions, or other serious medical conditions cannot participate. Furthermore, those who use medical devices, including but not limited to pacemakers, hearing aids, or defibrillators, or with a history of seizures, epilepsy, photosensitivity, severe dizziness, or loss of awareness triggered by visual stimuli cannot participate.

The participants are required to speak English well enough to understand the questionnaires and to take part in a semi-structured interview.

4. Do I have to take part?

It is up to you to decide whether or not to take part in this study. If you do decide to take part, you will be given this information sheet to keep and be asked to sign a consent form. You can withdraw at any time without giving a reason and without it affecting any benefits that you are entitled to. If you decide to withdraw you will be asked what you wish to happen to the data you have provided up to that point.

5. What will happen to me if I take part?

If you decide to take part:

1. You will be asked to sign a consent form to say that you agree to take part in the study. You can provide written consent prior to the study or orally on the day of the study if you like. You will be given a copy of this to keep for your records.
2. Once you sign a consent form the researcher will ask you to complete a questionnaire about your demographics (e.g., your age, gender, education, professional status), driving experience (e.g., license duration, driving frequency, kilometres driven last year), and opinions on autonomous driving or other mobility modes. You will have the option to leave a question blank or state 'prefer not answer' if you so choose. This information will be used to compare answers from people who are from different backgrounds.
3. Once you have completed the questionnaire, a researcher will give you a short presentation demonstrating the setup. After this presentation, you will drive in Virtual Reality for approximately 22 short rounds, rating the display after each round, followed by a short closing interview.

- The study will take place at UCL and will last for approximately 40 to 60 minutes
- The simulator will record your driving (speed, acceleration, steering, and the position of your vehicle), the resulting energy use, and your answers to the questionnaires.
- If you feel unwell, we will ask you to stop the study, remove the helmet and remain seated for a moment or accompany you to get some air if you experience any discomfort
- We provide breaks every 30 minutes or shorter, according to the manufacturer's usage descriptions

6. Will I be recorded and how will the recorded media be used?

You will not be audio- or video-recorded at any point. Only the driving data listed above, your questionnaire answers and written notes from the closing interview are recorded. No data will leave UCL premises.

7. What are the possible disadvantages and risks of taking part?

Virtual Reality environments can cause physical discomfort including nausea, eye strain, and disorientation. If this happens, we will abort the study.

8. What are the possible benefits of taking part?

There will be no immediate benefits to participating in the study. However, we hope to use the information we learn from this user study to make improvements to future mobility to make it more accessible and user-friendly.

Participation is unpaid: no payment or voucher is provided for taking part in this study.

9. What if something goes wrong?

If you have a concern about any aspect of this study, you should ask to speak with the researcher, Dr Mark Colley (email: m.colley@ucl.ac.uk) who will do his best to answer your questions. If your concerns are not yet resolved, you can contact the UCL Ethics committee at ethics@ucl.ac.uk.

10. Will my taking part in this project be kept confidential?

All information that is collected from you during the research will be kept strictly confidential and will be stored in accordance with the General Data Protection Regulation (GDPR) 2018. The only people in UCL who will have access to information that identifies you will be people who need to contact you to arrange the research appointments or audit the data collection process.

Your data will be held confidentially, with access restricted to the research team working on the project. The information we collect will be stored on computers at UCL in a form in which you cannot be identified. Only the study research team will have access to the data. The data are retained until they have been analysed and the study has ended. Only anonymized data will be made openly available via GitHub or similar sites.

11. Limits to confidentiality

Please note that assurances on confidentiality will be strictly adhered to unless evidence of wrongdoing or potential harm is uncovered. In such cases UCL may be obliged to contact relevant statutory bodies/agencies.

12. What will happen to the results of the research project?

We will publish the results in reports and papers and talk about the results at conferences, journals, seminars and teaching. Please tell the researchers if you would like a copy of any publications and we would be happy to send them to you when they are published. You will not be identified in any report/publication. With your permission, the data collected during the course of the project might be used for additional or subsequent research. Anonymised data will then be archived for 20 years in accordance with UCL guidelines.

13. Local Data Protection Privacy Notice

The controller for this project will be UCL. The UCL Data Protection Officer provides oversight of UCL activities involving the processing of personal data, and can be contacted at data-protection@ucl.ac.uk

This 'local' privacy notice sets out the information that applies to this particular study. Further information on how UCL uses participant information can be found in our 'general' privacy

notice: <https://www.ucl.ac.uk/legal-services/privacy/ucl-general-privacy-notice-participants-and-researchers-health-and-care-research-studies>

The information that is required to be provided to participants under data protection legislation (GDPR and DPA 2018) is provided across both the 'local' and 'general' privacy notices. The lawful basis that will be used to process your personal data are: "Public task" for personal data and "Research purposes" for special category data.

Your personal data will be processed so long as it is required for the research project. If we are able to anonymise or pseudonymise the personal data you provide we will undertake this, and will endeavour to minimise the processing of personal data wherever possible.

If you are concerned about how your personal data is being processed, or if you would like to contact us about your rights, please contact UCL in the first instance at data-protection@ucl.ac.uk.

Please do not mention your name or talk about private matters.

14. Contact for further information

Dr Mark Colley
UCL Interaction Centre
m.colley@ucl.ac.uk
66-72 Gower St
London WC1E 6EA

Thank you for reading this information sheet and for considering to take part in this research study. *You will be given a copy of this information sheet and a signed consent form to keep.*

G.3 Consent Form

The original consent form (three pages) is reproduced on the following pages.

CONSENT FORM FOR ADULTS IN RESEARCH STUDIES

Please complete this form after you have read the Information Sheet and/or listened to an explanation about the research.

Title of Study: Sustainable Personalised Driving

Department: UCL Interaction Centre

Name and Contact Details of the Researcher(s):

Yike Zhang Email: ucab338@ucl.ac.uk

Name and Contact Details of the Principal Researcher:

Mark Colley Email: m.colley@ucl.ac.uk

Name and Contact Details of the UCL Data Protection Officer:

Alexandra Potts Email: data-protection@ucl.ac.uk

This study has been approved by the UCL Research Ethics Committee: Project ID number: 1165

Thank you for considering taking part in this research. The person organising the research must explain the project to you before you agree to take part. If you have any questions arising from the Information Sheet or explanation already given to you, please ask the researcher before you decide whether to join in. You will be given a copy of this Consent Form to keep and refer to at any time.

I confirm that I understand that by ticking/initialling each box below I am consenting to this element of the study. I understand that it will be assumed that unticked/initialled boxes means that I DO NOT consent to that part of the study. I understand that by not giving consent for any one element that I may be deemed ineligible for the study.

		Tick Box
1.	*I confirm that I have read and understood the Information Sheet for the above study. I have had an opportunity to consider the information and what will be expected of me. I have also had the opportunity to ask questions which have been answered to my satisfaction	
2.	*I consent to participate in the study. I understand that my personal information including, my name, age, gender, email address and/or telephone number, will be used for the purposes explained to me. I understand that according to data protection legislation, 'public task' will be the lawful basis for processing and 'research purposes' will be the lawful basis for processing special category data.	
3.	Use of the information for this project only * I understand that my personal information will remain confidential as far as possible, and that all efforts will be made to ensure I cannot be identified. I understand that my data gathered in this study will be stored pseudo-anonymously and securely in accordance with General Data Protection Regulation (GDPR) and the Data Protection Act 2018., and only the researchers involved in the research project will have access to the data. Derivations from this are highlighted in the next entries. It will not be possible to identify me in any publications.	
4.	*I understand that my information may be subject to review by responsible individuals from the University College London (the study sponsor) for monitoring and audit purposes.	
5.	*I understand that my participation is voluntary and that I am free to withdraw at any time during the user study without giving a reason.	

6.	I understand the potential risks of participating and the support that will be available to me should I become distressed during the course of the research.	Zhang
7.	I understand the direct/indirect benefits for wider society of participating.	
8.	I understand that the data (beyond the anonymized questionnaires) will not be made available to any commercial organisations but is solely the responsibility of the researcher(s) undertaking this study.	
9.	I understand that I will not benefit financially from this study or from any possible outcome it may result in in the future.	
10.	I understand that participation in this study is unpaid, and that no compensation is provided, including if I choose to withdraw.	
11.	I agree that my pseudonymised research data (i.e., the driving data recorded by the simulator and my questionnaire responses) may be used by others for future research. No one will be able to identify you when this data is shared.	
12.	I understand that the information I have submitted will be published as a report. I wish to receive a copy of it.	☐ Yes ☐ No ☐ Yes ☐ No
13.	I hereby confirm that I understand the inclusion criteria as detailed in the Information Sheet and explained to me by the researcher.	
14.	I hereby confirm that: (a) I understand the exclusion criteria as detailed in the Information Sheet (i.e., no dementia and severe illnesses; not pregnant, elderly, or have pre-existing binocular vision abnormalities, psychiatric disorders, heart conditions, or other serious medical conditions cannot participate; not using medical devices, including but not limited to pacemakers, hearing aids, or defibrillators, or with a history of seizures, epilepsy, photosensitivity, severe dizziness, or loss of awareness triggered by visual stimuli) and explained to me by the researcher; and (b) I do not fall under the exclusion criteria.	
15.	I am aware of who I should contact if I wish to lodge a complaint.	
16.	I voluntarily agree to take part in this study.	
17.	Use of information for this project and beyond I would be happy for the data I provide to be archived at UCL. I understand that other authenticated researchers will have access to my pseudonymised data.	
18.	I understand that the researcher will not pass to any third party any of the personal information I provided them with. However, I understand that I should not provide any personal information (e.g., names or addresses) or sensitive personal account (e.g., information about an individual's racial or ethnic origin, political opinions, religious beliefs, trade union membership) of myself or others to limit the possibility to be identified by third parties.	

If you would like your contact details to be retained so that you can be contacted in the future by UCL researchers who would like to invite you to participate in follow up studies to this project, or in future studies of a similar nature, please tick the appropriate box below.

☐	Yes, I would be happy to be contacted in this way	
☐	No, I would not like to be contacted	

Name of participant

Date

Signature

Researcher

Date

Signature